Software Requirement Specification

for

Arena 3

Sports Centre Management System

Version 2.0

Prepared by
Arena3 team

October 2026

Table of Contents

Record of changes 4

I. Overall requirements 5

1. Product overview 5
2. Context diagram 6
3. Main workflows 8
   - 3.1 Register a member account 8
   - 3.2 Book a court and pay 10
   - 3.3 Enrol in a class or join the waitlist 12
   - 3.4 Refund a payment 14

II. User requirements 16

1. Actors 16
2. Use case diagrams 18
   - 2.1 Account and access 18
   - 2.2 Member: courts, plans and payment 19
   - 2.3 Member: classes and training 20
   - 2.4 Receptionist: sales and the till 21
   - 2.5 Receptionist: daily operations 22
   - 2.6 Coach 23
   - 2.7 Manager: setup 24
   - 2.8 Manager: oversight 25
3. Use case list 26

III. Functional requirements 29

1. System functional overview 29
   - 1.1 Screens flow 29
   - 1.2 Screen descriptions 34
   - 1.3 Screen authorization 35
   - 1.4 Non-screen functions 36
   - 1.5 Entity relationship diagram 37
   - 1.6 Class diagrams 39
2. Use case specifications 46
   - 2.1 UC-02 Register a member account 46
   - 2.2 UC-12 Book a court and pay online 49
   - 2.3 UC-24 Enrol in a class 52
   - 2.4 UC-46 Check a member in at the gate 53
   - 2.5 UC-43 Raise a refund 55

IV. Non-functional requirements 57

1. External interfaces 57
   - 1.1 User interfaces 57
   - 1.2 Software interfaces 57

1.3 API error codes 58
2. Quality attributes 58
2.1 Correctness and reliability 58
2.2 Performance 58
2.3 Usability 59
2.4 Auditability 59
2.5 Security and privacy 59
2.6 Maintainability 59
V. Requirement appendix 60
1. Business rules 60
2. Centre settings 61
3. Common requirements 62
4. Application messages 62

List of Figures

Figure 1. Context diagram of the Arena3 Sports Centre Management System 6
Figure 2. Activity diagram: Register a member account 8
Figure 3. Activity diagram: Book a court and pay 10
Figure 4. Activity diagram: Enrol in a class or join the waitlist 12
Figure 5. Activity diagram: Refund a payment 14
Figure 6. Actor generalisation and system actors 16
Figure 7. Use case diagram: Account and access 18
Figure 8. Use case diagram: Member court booking and payment 19
Figure 9. Use case diagram: Member classes and training 20
Figure 10. Use case diagram: Front desk sales 21
Figure 11. Use case diagram: Front desk operations 22
Figure 12. Use case diagram: Coach 23
Figure 13. Use case diagram: Manager setup 24
Figure 14. Use case diagram: Manager oversight 25
Figure 15. Screens flow: public site and sign-in 29
Figure 16. Screens flow: member portal 30
Figure 17. Screens flow: front desk (receptionist) 31
Figure 18. Screens flow: coach portal 32
Figure 19. Screens flow: manager portal 33
Figure 20. Conceptual entity relationship diagram 37
Figure 21. Class diagram: booking and payment 39
Figure 22. Class diagram: membership and classes 41
Figure 23. Class diagram: coaching and training 43
Figure 24. Class diagram: centre operations 45
Figure 25. Sequence diagram: register a member account 48
Figure 26. Sequence diagram: book a court and pay by payOS QR 51
Figure 27. Sequence diagram: check a member in at the gate 54
Figure 28. Sequence diagram: raise and approve a refund 56

Record of changes

This record lists every released version of the SRS, newest first. A = added, M = modified, D = deleted.

Date	Version	A/M/D	In charge	Description
2026-10-11	2.0	M	Arena3 team	Redrew all 28 UML figures to the FPT template and Learning UML 2.0 rules. Aligned every section with the deployed app.
2026-10-04	1.4.1	A, D	Arena3 team	Added QR gate check-in, promo codes, at-risk list, deposits, fixed weekly bookings, day passes and loyalty points. Removed member debt and court conversion.
2026-09-27	1.4	M	Arena3 team	Traced each requirement to its endpoint or business rule. Added online payment by payOS.
2026-09-20	1.1	A	Arena3 team	Added receipts, customer care requests and the training module (feature flag F4).

I. Overall requirements

1. Product overview

Arena3 is one web application that runs a single sports centre in Ho Chi Minh City, from court booking to the end-of-day till. It serves badminton, basketball and volleyball courts, membership plans, coached classes and the front desk.

Portal	Route	Who uses it	Main purpose
Public site	`/`, `/login`, `/register`	Guest	Browse prices and classes, sign up, sign in
Member portal	`/app`	Member	Book courts, pay, enrol in classes, show the gate pass
Front desk	`/desk`	Receptionist	Serve walk-ins, take payments, run the gate and the till
Coach portal	`/coach`	Coach	Teach classes, take attendance, track student progress
Manager portal	`/manager`	Manager	Set prices and rules, staff the centre, read reports

The app is built with TanStack Start and React, with a JSON API under `/v1` in the same deployment. Production runs on Vercel with Neon PostgreSQL. Development runs offline on an embedded PGLite database that migrates and seeds itself.

Out of scope. Multi-branch operation, payroll, native mobile apps and SMS delivery are not part of this release. Class fees and gear rental are collected outside the payment ledger.

2. Context diagram

The context diagram shows Arena3 as one process exchanging data with five human actors and three external systems.

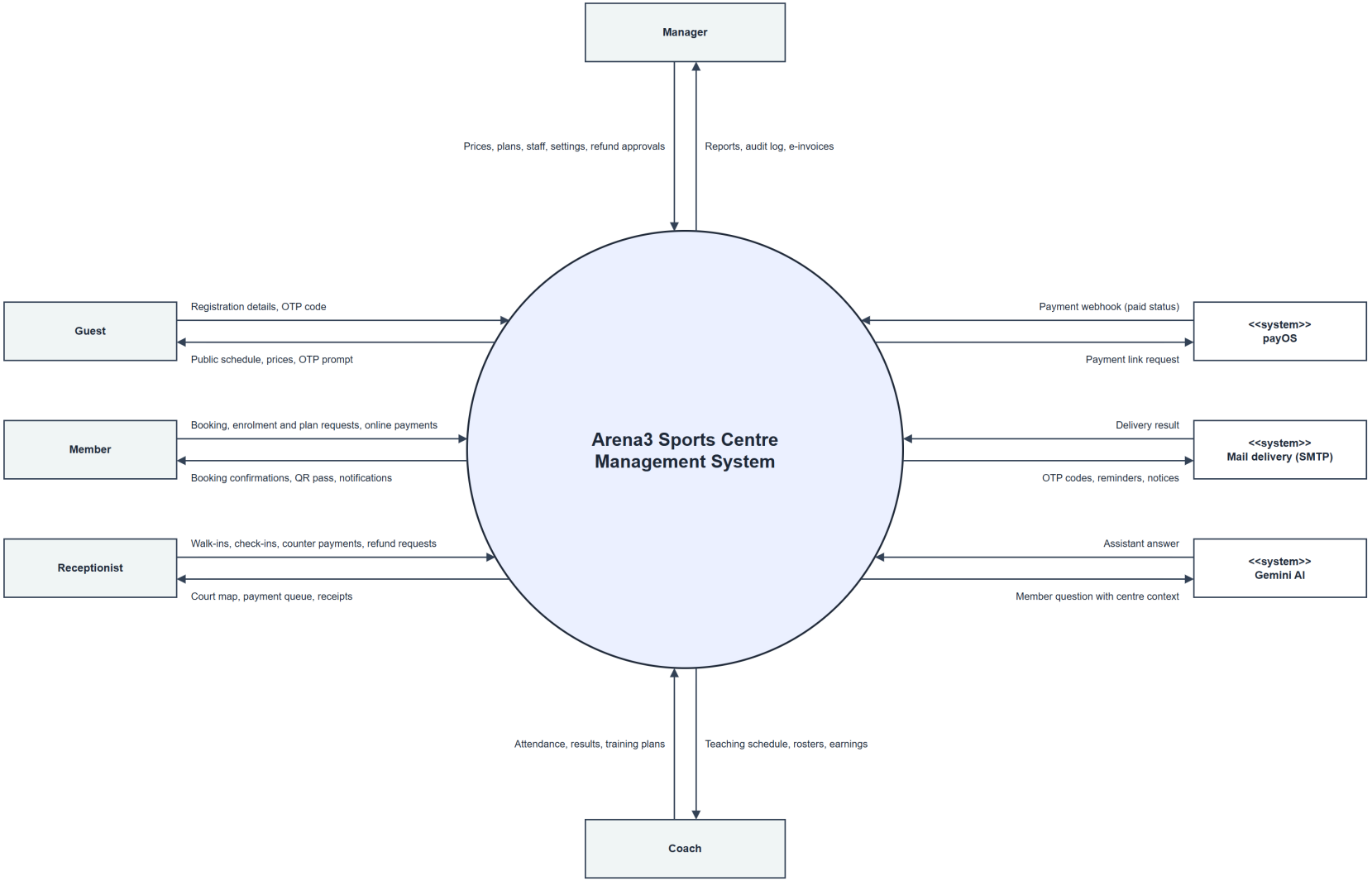


Figure 1. *Context diagram of the Arena3 Sports Centre Management System*

External entity	Data into Arena3	Data out of Arena3
Guest	Registration details, OTP code	Public schedule, prices, OTP prompt
Member	Booking, enrolment and plan requests, online payments	Booking confirmations, QR pass, notifications
Receptionist	Walk-ins, check-ins, counter payments, refund requests	Court map, payment queue, receipts
Coach	Attendance, results, training plans	Teaching schedule, rosters, earnings
Manager	Prices, plans, staff, settings, refund approvals	Reports, audit log, e-invoices
payOS	Signed "paid" webhook	Payment link request
Mail delivery (SMTP)	Delivery result	OTP codes, reminders, notices
Gemini AI	Assistant answer	Member question with centre context only

3. Main workflows

Four activity diagrams cover the workflows that move money or change a schedule. Each uses swimlanes, one initial node, guarded decisions and one final node.

3.1 Register a member account

A guest becomes a member only after proving the email address with a one-time code.

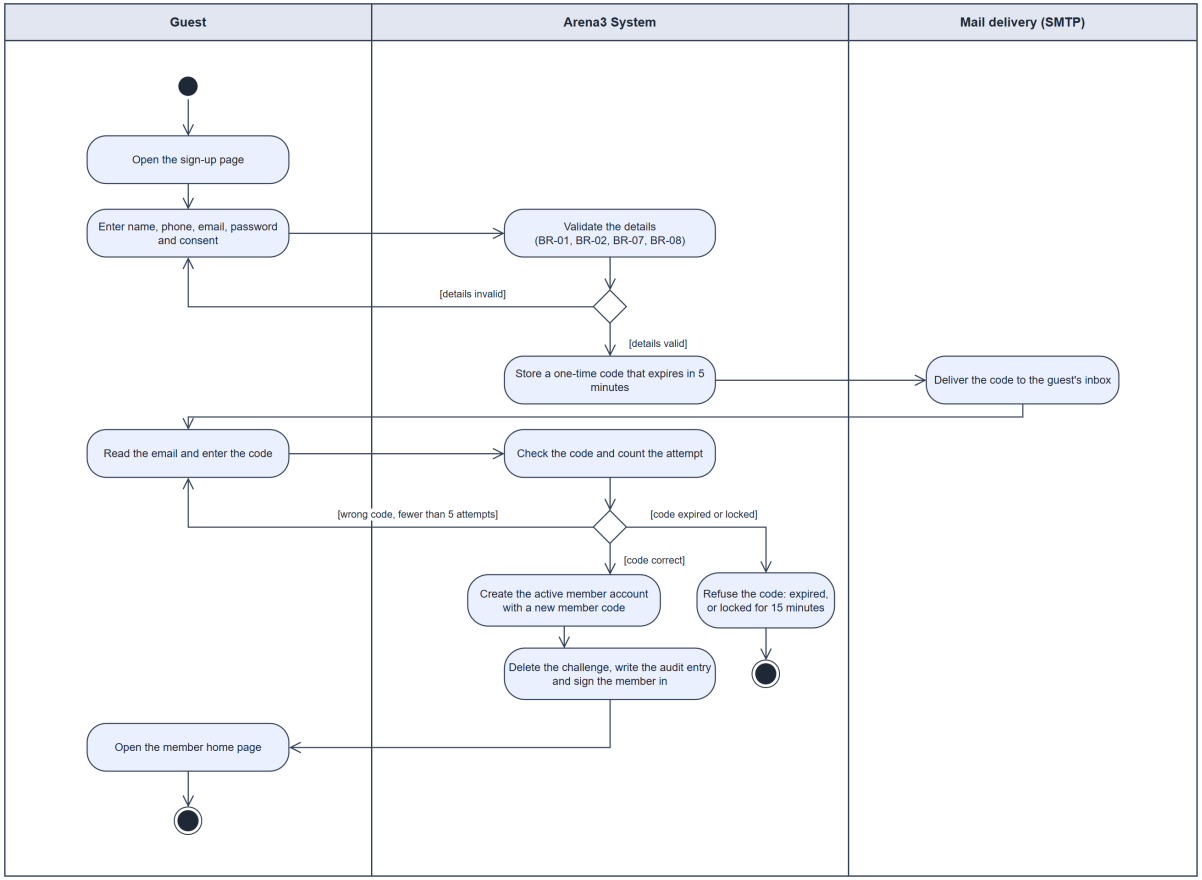


Figure 2. *Activity diagram: Register a member account*

The code expires after 5 minutes. Too many wrong attempts lock the challenge for 15 minutes. A successful check creates the account, writes an audit entry and signs the member in.

3.2 Book a court and pay

A booking is confirmed only when money is known to have arrived or plan hours are spent.

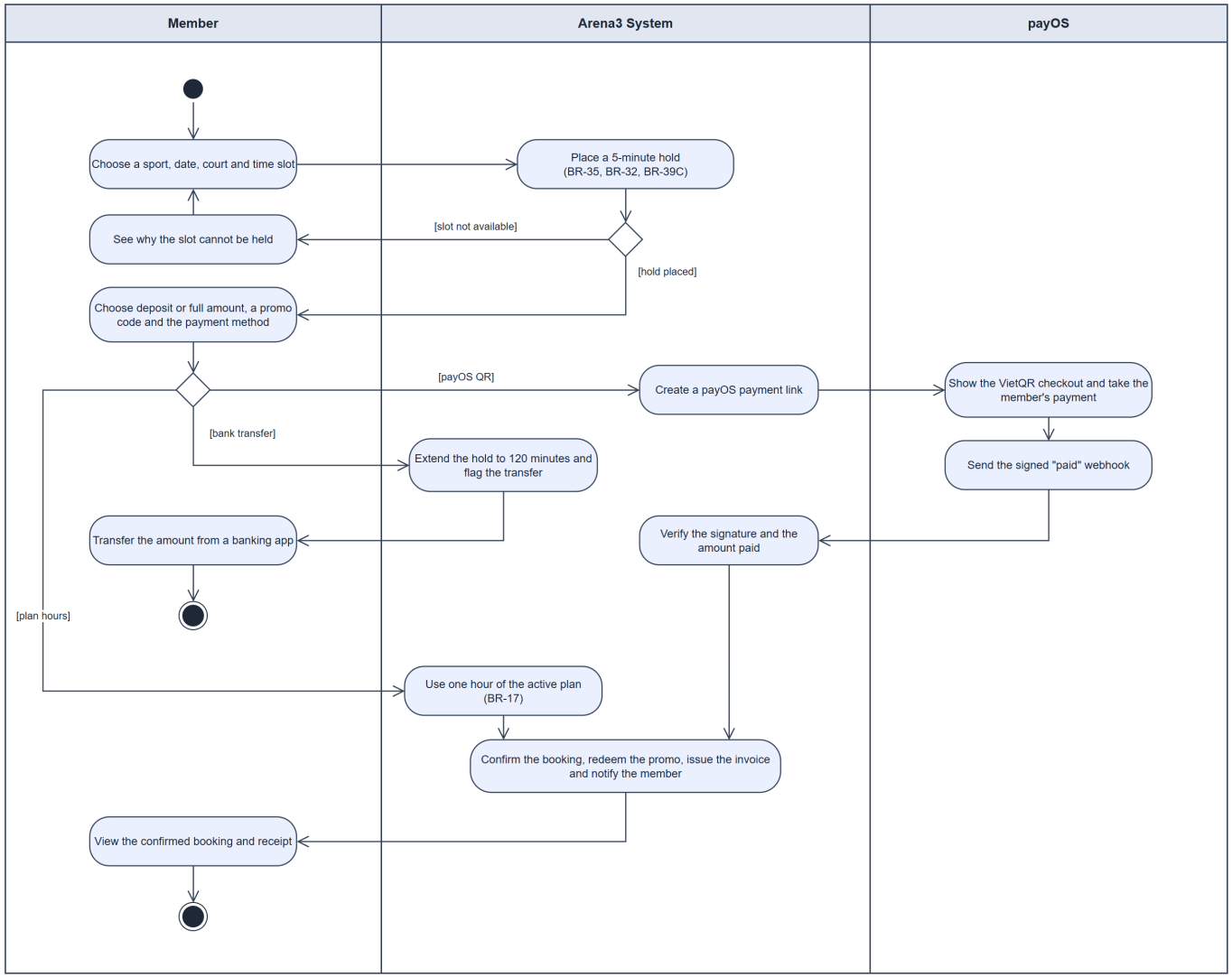


Figure 3. *Activity diagram: Book a court and pay*

A slot is held for 5 minutes while the member pays. Three paths follow:

- **payOS QR:** the booking is confirmed only after Arena3 verifies payOS's signed result and the amount.
- **Bank transfer:** the hold stretches to 120 minutes and the transfer waits for the front desk to reconcile it.
- **Plan quota:** one court hour is taken from the member's active plan.

3.3 Enrol in a class or join the waitlist

A full class never refuses a member outright; it offers a waitlist place instead.

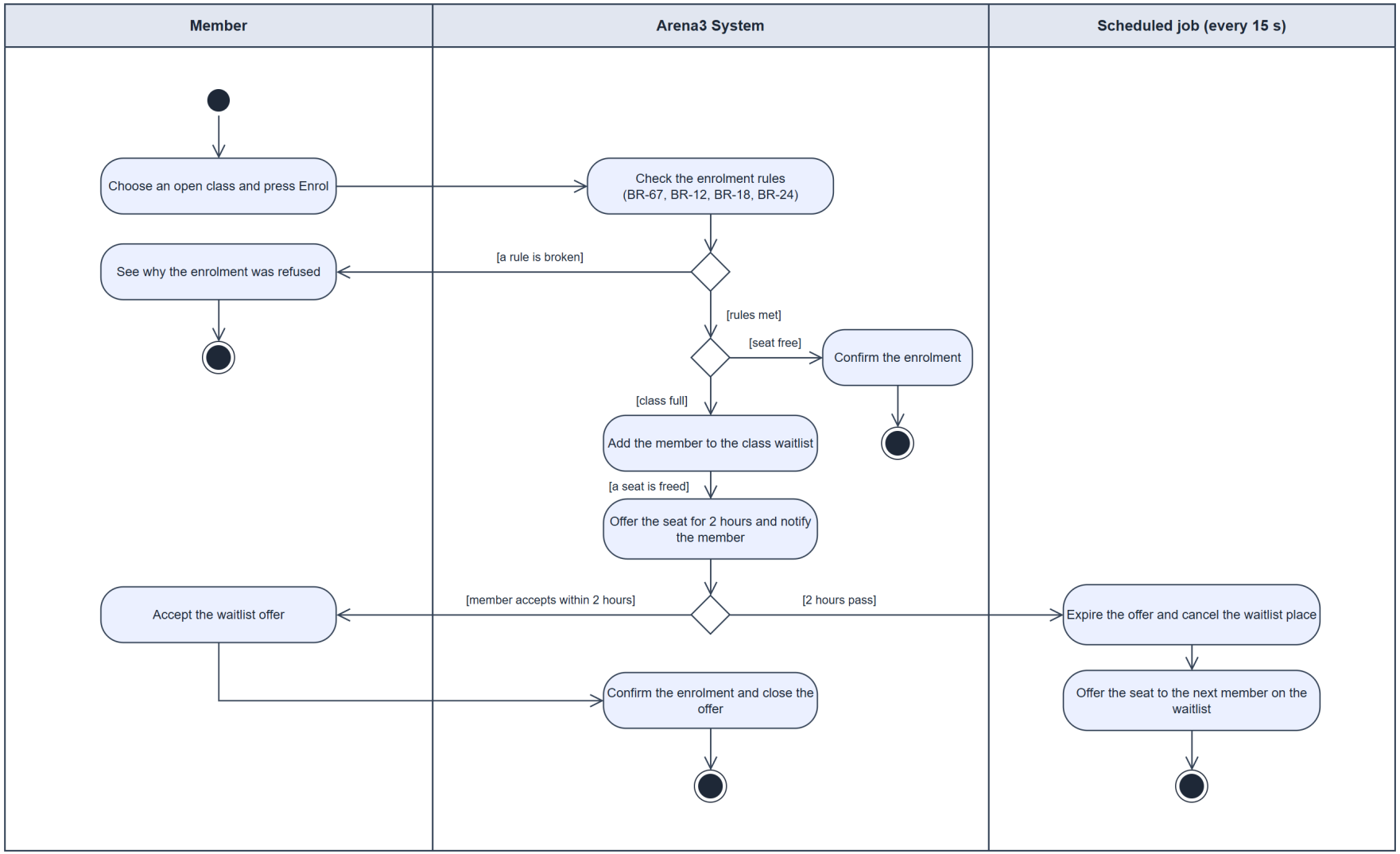


Figure 4. *Activity diagram: Enrol in a class or join the waitlist*

When a seat frees up, the next member gets a 2-hour offer. The scheduled job expires an unanswered offer and moves on to the next member.

3.4 Refund a payment

A large refund needs a second person, so no receptionist can refund their own sale above the limit.

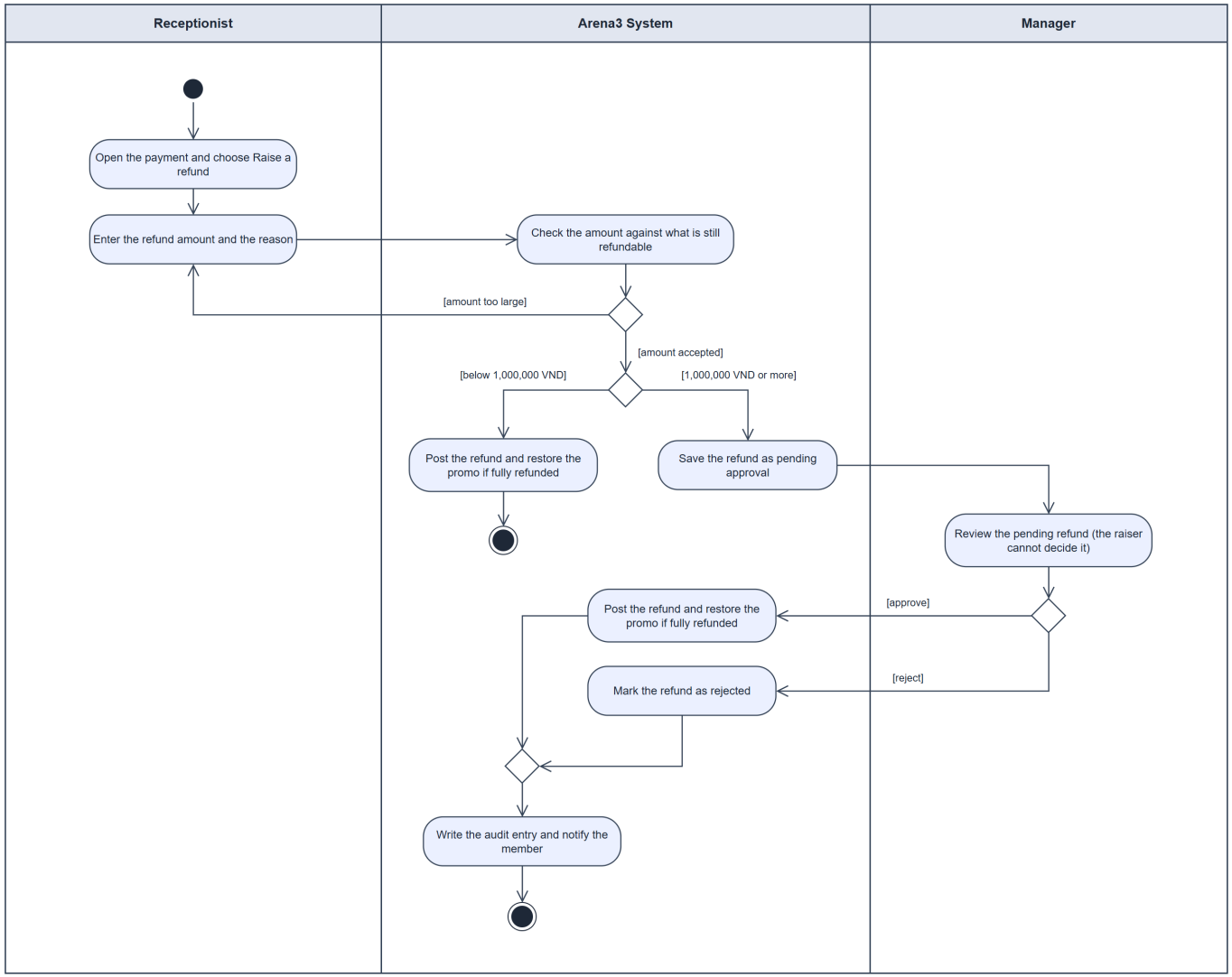


Figure 5. *Activity diagram: Refund a payment*

Refunds up to 1,000,000đ post at once. Larger refunds wait for a manager, who approves or rejects them. Both outcomes are audited and the member is notified.

II. User requirements

1. Actors

Arena3 has five human actors and three external system actors. The four staff and member roles specialise one abstract Registered User, who can sign in.

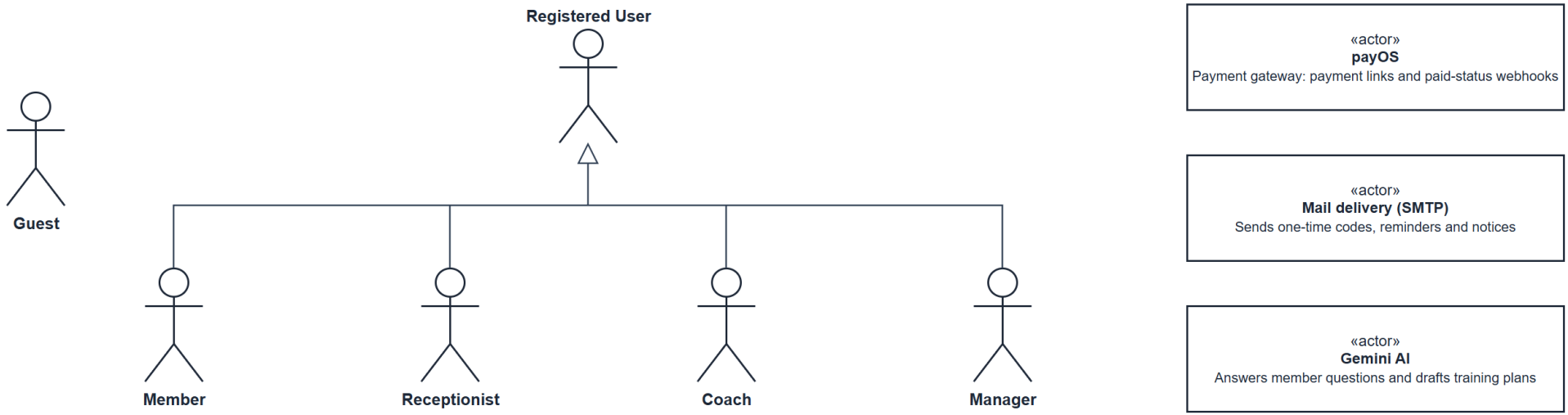


Figure 6. *Actor generalisation and system actors*

#	Actor	Kind	Description
1	Guest	Human	Anyone not signed in. Browses prices and classes and registers a member account.
2	Registered User	Abstract	Any signed-in person. Owns the sign-in, profile, password and notification use cases.
3	Member	Human	A customer with an account. Books courts, buys plans, joins classes and enters with a QR pass.
4	Receptionist	Staff	Runs the front desk and the till: walk-ins, payments, gate, gear, refunds up to the limit.
5	Coach	Staff	Teaches classes. Takes attendance and records results, plans, homework and reviews.
6	Manager	Staff	Sets prices, plans, classes and rules. Issues staff accounts, approves large refunds and reads reports.
7	payOS	System	Vietnamese payment gateway. Issues VietQR payment links and sends signed "paid" webhooks.
8	Mail delivery (SMTP)	System	Sends one-time codes, reminders and notices by email.
9	Gemini AI	System	Google's model. Answers member questions and drafts training plans from centre data only.

Staff accounts are issued by a manager; there is no public staff sign-up and no admin role.

2. Use case diagrams

Each diagram covers one feature area with one system boundary. Associations carry no arrowheads; «include» points from the base case and «extend» points to it.

2.1 Account and access

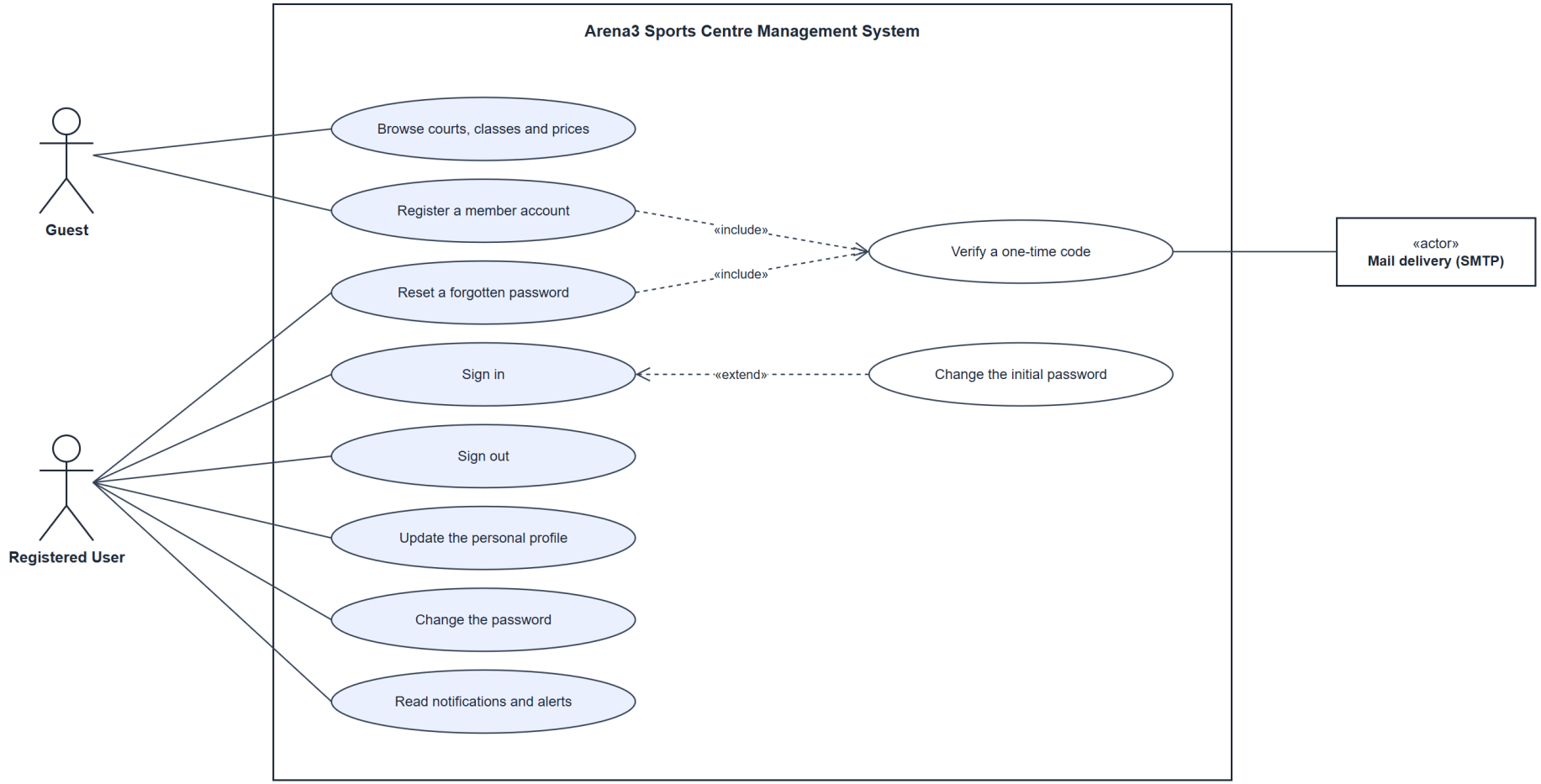


Figure 7. *Use case diagram: Account and access*

2.2 Member: courts, plans and payment

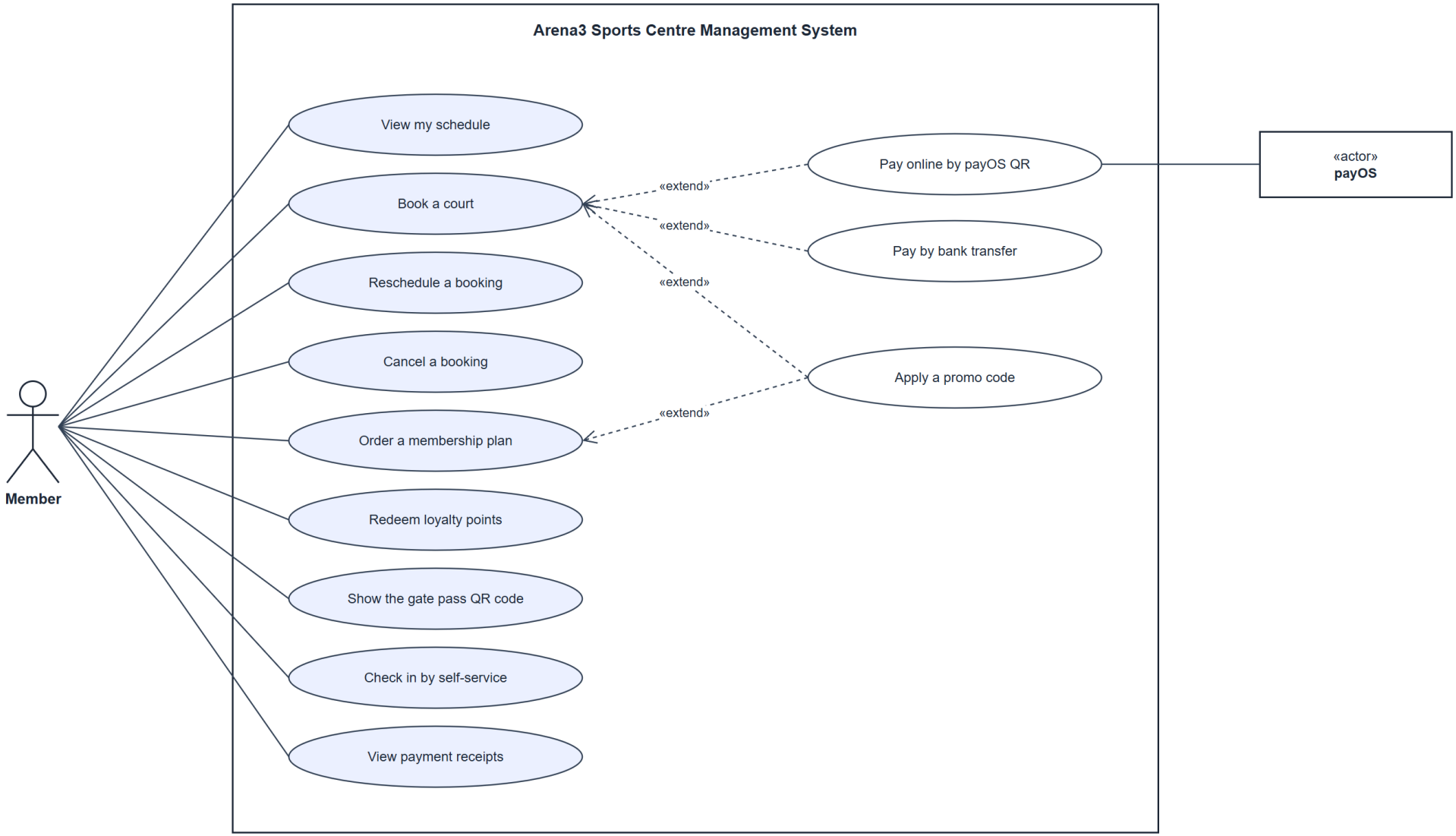


Figure 8. *Use case diagram: Member court booking and payment*

2.3 Member: classes and training

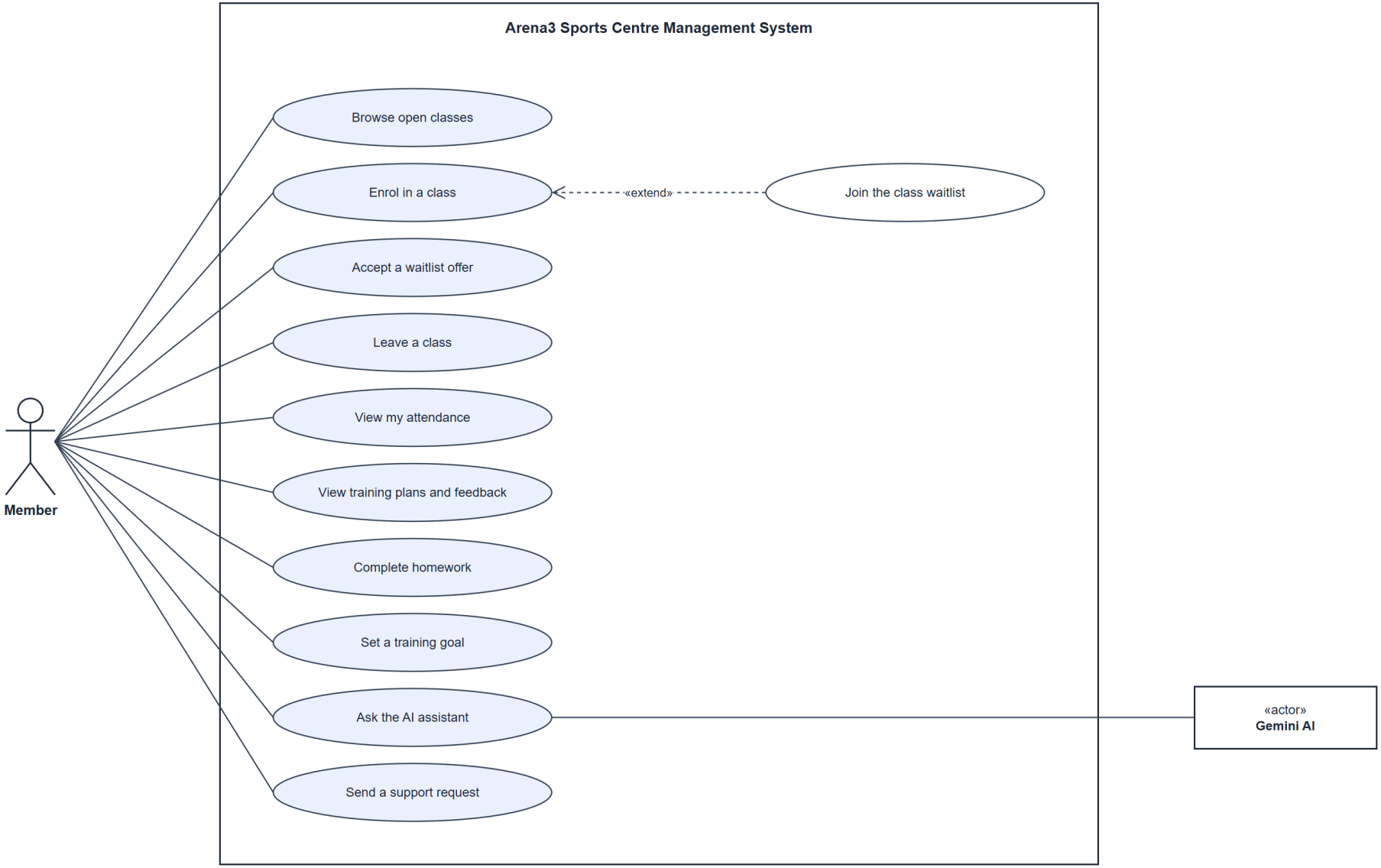


Figure 9. *Use case diagram: Member classes and training*

2.4 Receptionist: sales and the till

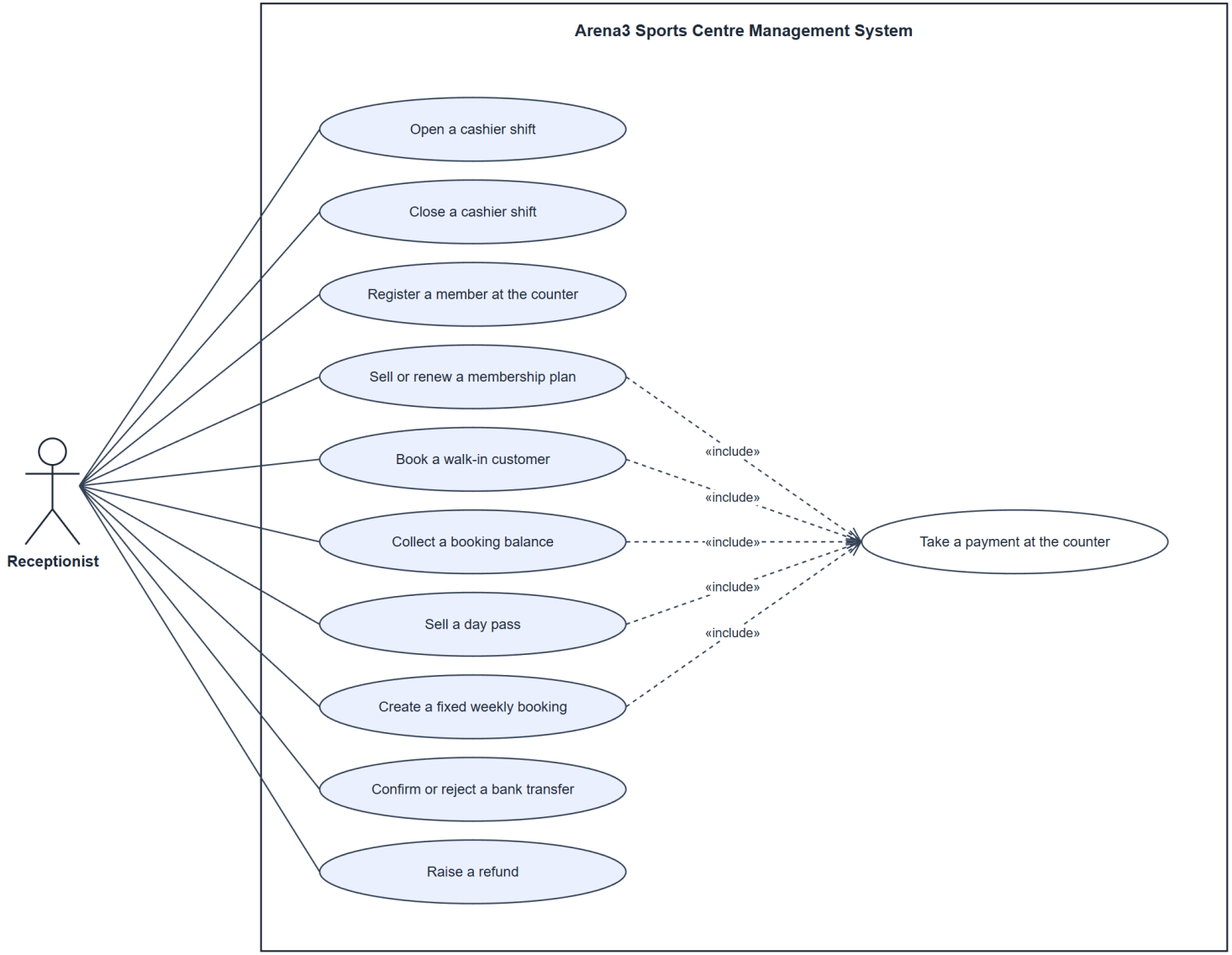


Figure 10. *Use case diagram: Front desk sales*

2.5 Receptionist: daily operations

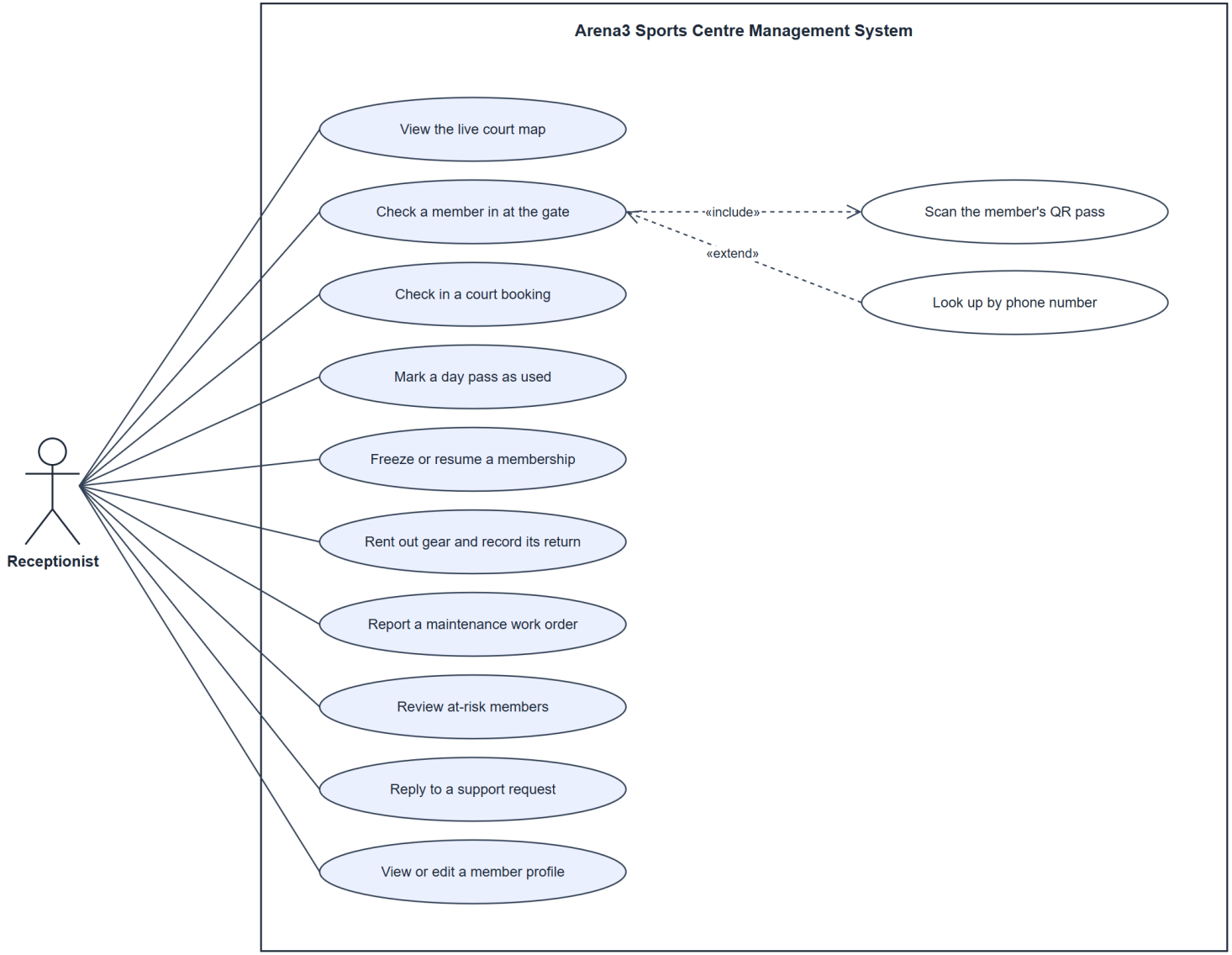


Figure 11. *Use case diagram: Front desk operations*

2.6 Coach

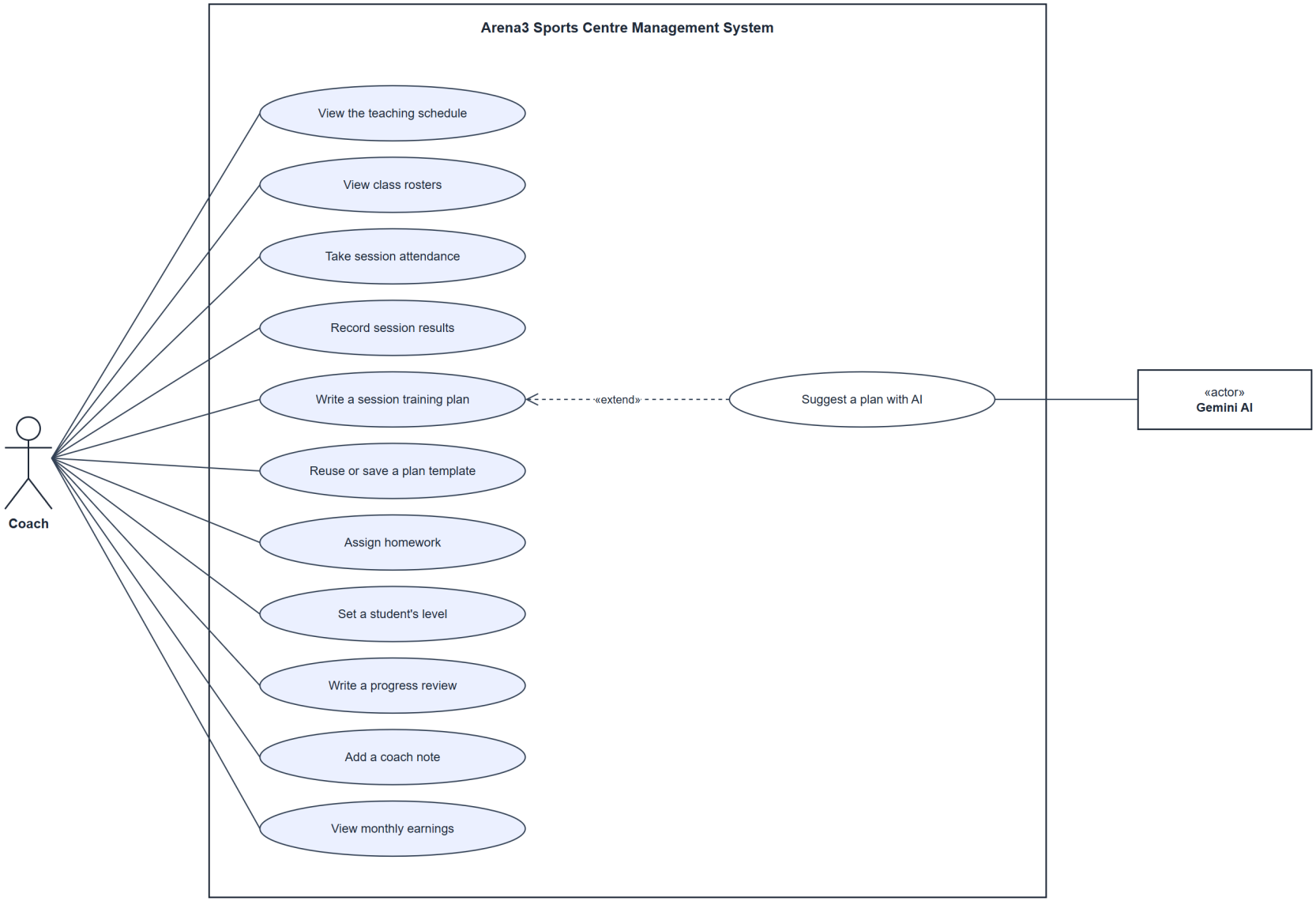


Figure 12. *Use case diagram: Coach*

2.7 Manager: setup

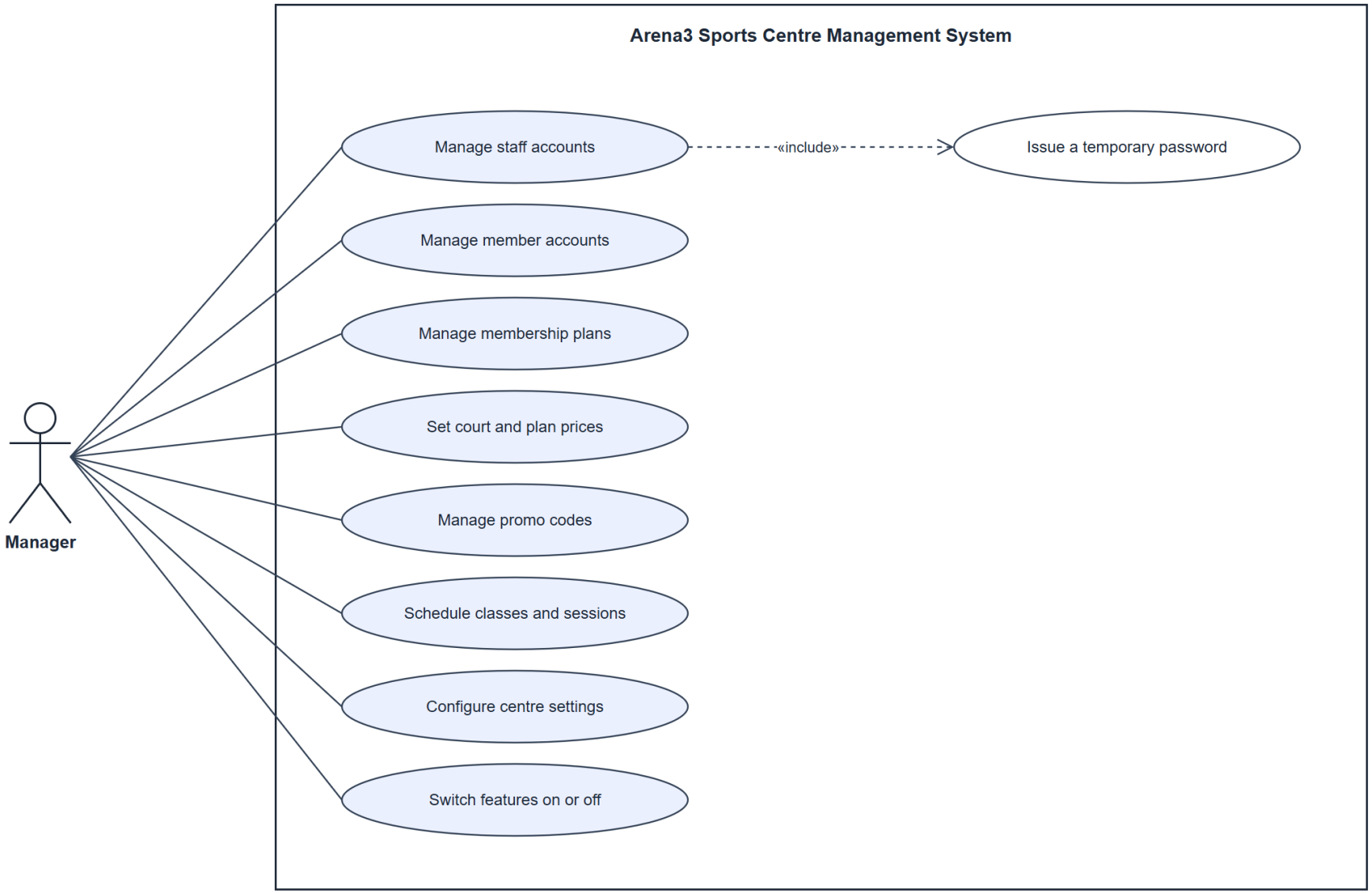


Figure 13. *Use case diagram: Manager setup*

2.8 Manager: oversight

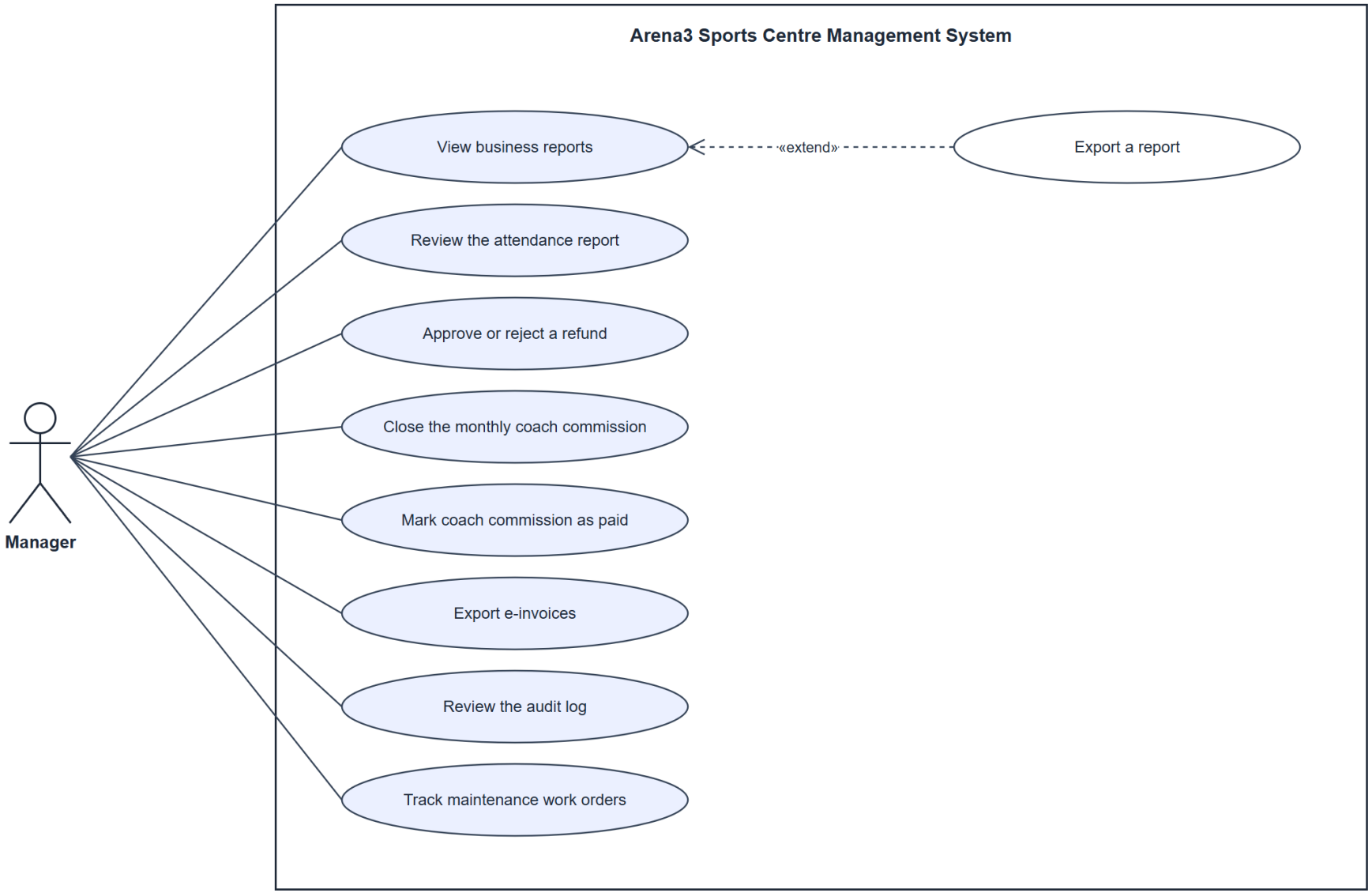


Figure 14. *Use case diagram: Manager oversight*

3. Use case list

The list numbers all 86 use cases in figure order. Cases marked «include» or «extend» only run as part of the base case named in their figure.

ID	Use case	Primary actor	Figure
UC-01	Browse courts, classes and prices	Guest	7
UC-02	Register a member account	Guest	7
UC-03	Reset a forgotten password	Registered User	7
UC-04	Sign in	Registered User	7
UC-05	Sign out	Registered User	7
UC-06	Update the personal profile	Registered User	7
UC-07	Change the password	Registered User	7
UC-08	Read notifications and alerts	Registered User	7
UC-09	Verify a one-time code «include»	Guest, Registered User	7
UC-10	Change the initial password «extend»	Registered User	7
UC-11	View my schedule	Member	8
UC-12	Book a court	Member	8
UC-13	Reschedule a booking	Member	8
UC-14	Cancel a booking	Member	8
UC-15	Order a membership plan	Member	8
UC-16	Redeem loyalty points	Member	8
UC-17	Show the gate pass QR code	Member	8
UC-18	Check in by self-service	Member	8
UC-19	View payment receipts	Member	8
UC-20	Pay online by payOS QR «extend»	Member	8
UC-21	Pay by bank transfer «extend»	Member	8
UC-22	Apply a promo code «extend»	Member	8
UC-23	Browse open classes	Member	9
UC-24	Enrol in a class	Member	9
UC-25	Accept a waitlist offer	Member	9
UC-26	Leave a class	Member	9
UC-27	View my attendance	Member	9
UC-28	View training plans and feedback	Member	9
UC-29	Complete homework	Member	9
UC-30	Set a training goal	Member	9
UC-31	Ask the AI assistant	Member	9
UC-32	Send a support request	Member	9
UC-33	Join the class waitlist «extend»	Member	9
UC-34	Open a cashier shift	Receptionist	10

ID	Use case	Primary actor	Figure
UC-35	Close a cashier shift	Receptionist	10
UC-36	Register a member at the counter	Receptionist	10
UC-37	Sell or renew a membership plan	Receptionist	10
UC-38	Book a walk-in customer	Receptionist	10
UC-39	Collect a booking balance	Receptionist	10
UC-40	Sell a day pass	Receptionist	10
UC-41	Create a fixed weekly booking	Receptionist	10
UC-42	Confirm or reject a bank transfer	Receptionist	10
UC-43	Raise a refund	Receptionist	10
UC-44	Take a payment at the counter «include»	Receptionist	10
UC-45	View the live court map	Receptionist	11
UC-46	Check a member in at the gate	Receptionist	11
UC-47	Check in a court booking	Receptionist	11
UC-48	Mark a day pass as used	Receptionist	11
UC-49	Freeze or resume a membership	Receptionist	11
UC-50	Rent out gear and record its return	Receptionist	11
UC-51	Report a maintenance work order	Receptionist	11
UC-52	Review at-risk members	Receptionist	11
UC-53	Reply to a support request	Receptionist	11
UC-54	View or edit a member profile	Receptionist	11
UC-55	Scan the member's QR pass «include»	Receptionist	11
UC-56	Look up by phone number «extend»	Receptionist	11
UC-57	View the teaching schedule	Coach	12
UC-58	View class rosters	Coach	12
UC-59	Take session attendance	Coach	12
UC-60	Record session results	Coach	12
UC-61	Write a session training plan	Coach	12
UC-62	Reuse or save a plan template	Coach	12
UC-63	Assign homework	Coach	12
UC-64	Set a student's level	Coach	12
UC-65	Write a progress review	Coach	12
UC-66	Add a coach note	Coach	12
UC-67	View monthly earnings	Coach	12
UC-68	Suggest a plan with AI «extend»	Coach	12
UC-69	Manage staff accounts	Manager	13
UC-70	Manage member accounts	Manager	13
UC-71	Manage membership plans	Manager	13
UC-72	Set court and plan prices	Manager	13
UC-73	Manage promo codes	Manager	13

ID	Use case	Primary actor	Figure
UC-74	Schedule classes and sessions	Manager	13
UC-75	Configure centre settings	Manager	13
UC-76	Switch features on or off	Manager	13
UC-77	Issue a temporary password «include»	Manager	13
UC-78	View business reports	Manager	14
UC-79	Review the attendance report	Manager	14
UC-80	Approve or reject a refund	Manager	14
UC-81	Close the monthly coach commission	Manager	14
UC-82	Mark coach commission as paid	Manager	14
UC-83	Export e-invoices	Manager	14
UC-84	Review the audit log	Manager	14
UC-85	Track maintenance work orders	Manager	14
UC-86	Export a report «extend»	Manager	14

III. Functional requirements

1. System functional overview

1.1 Screens flow

Each portal has one screen flow that starts at its entry point and reads left to right. Solid boxes are routed screens; dashed boxes are dialogs or in-page steps; grey boxes are portal homes drawn in their own flow.

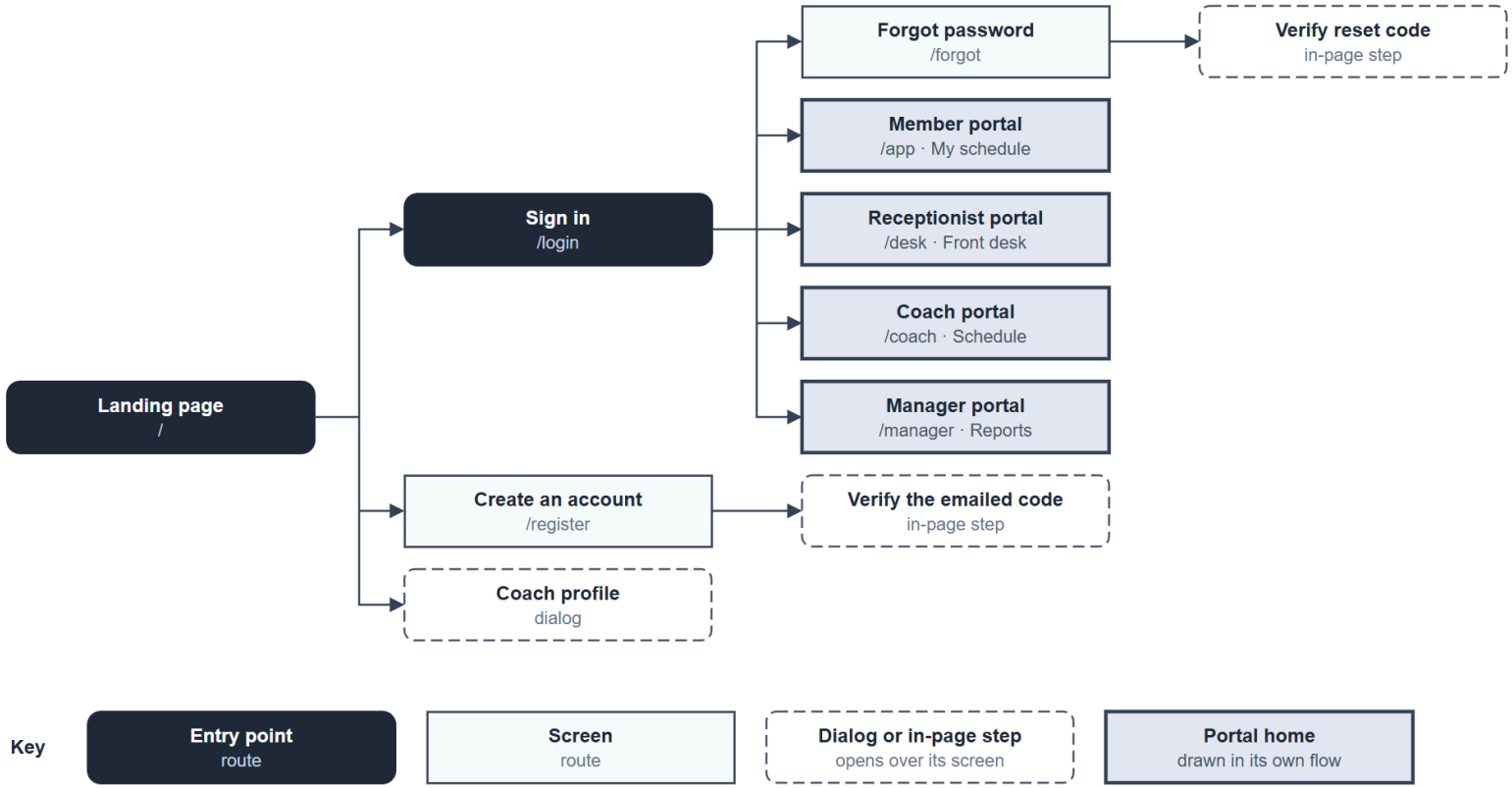


Figure 15. *Screens flow: public site and sign-in*

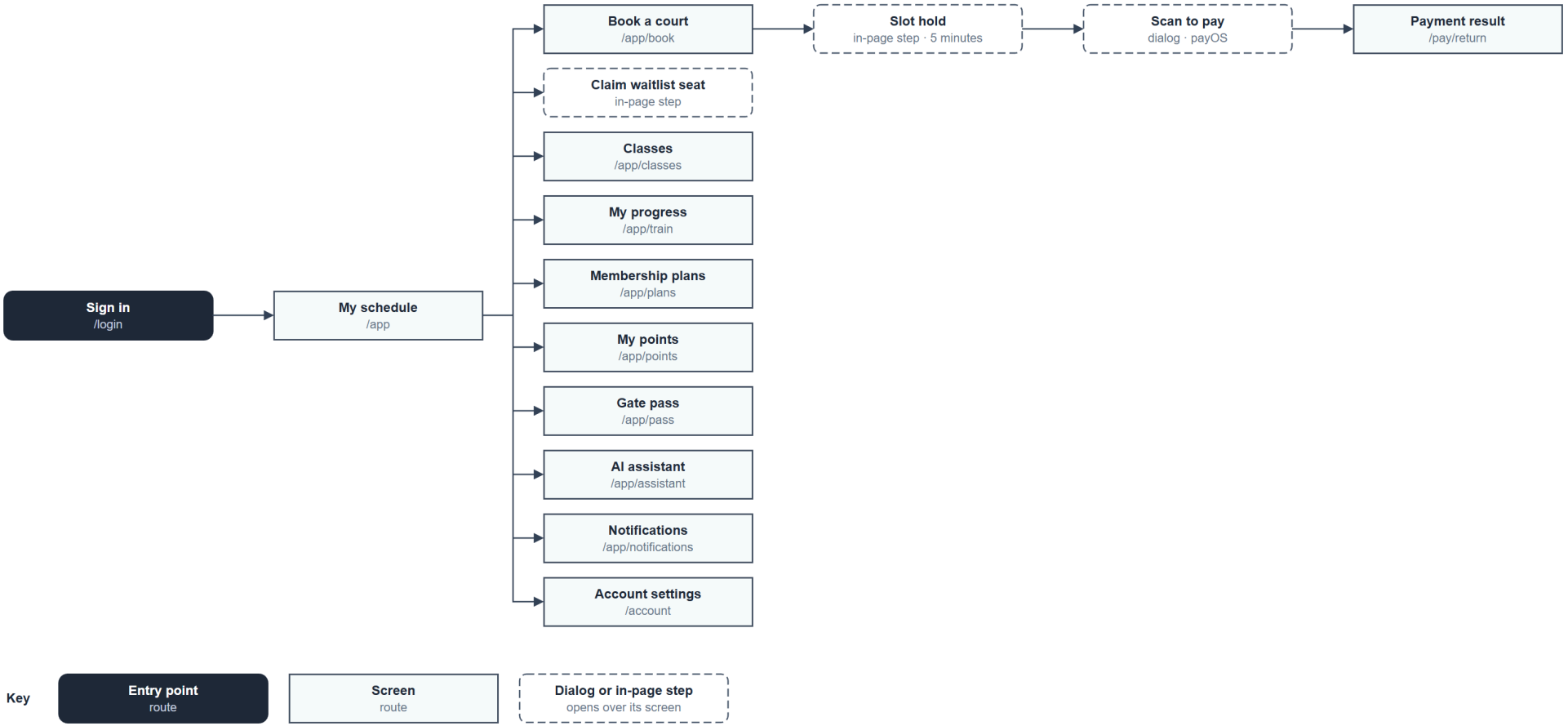


Figure 16. *Screens flow: member portal*

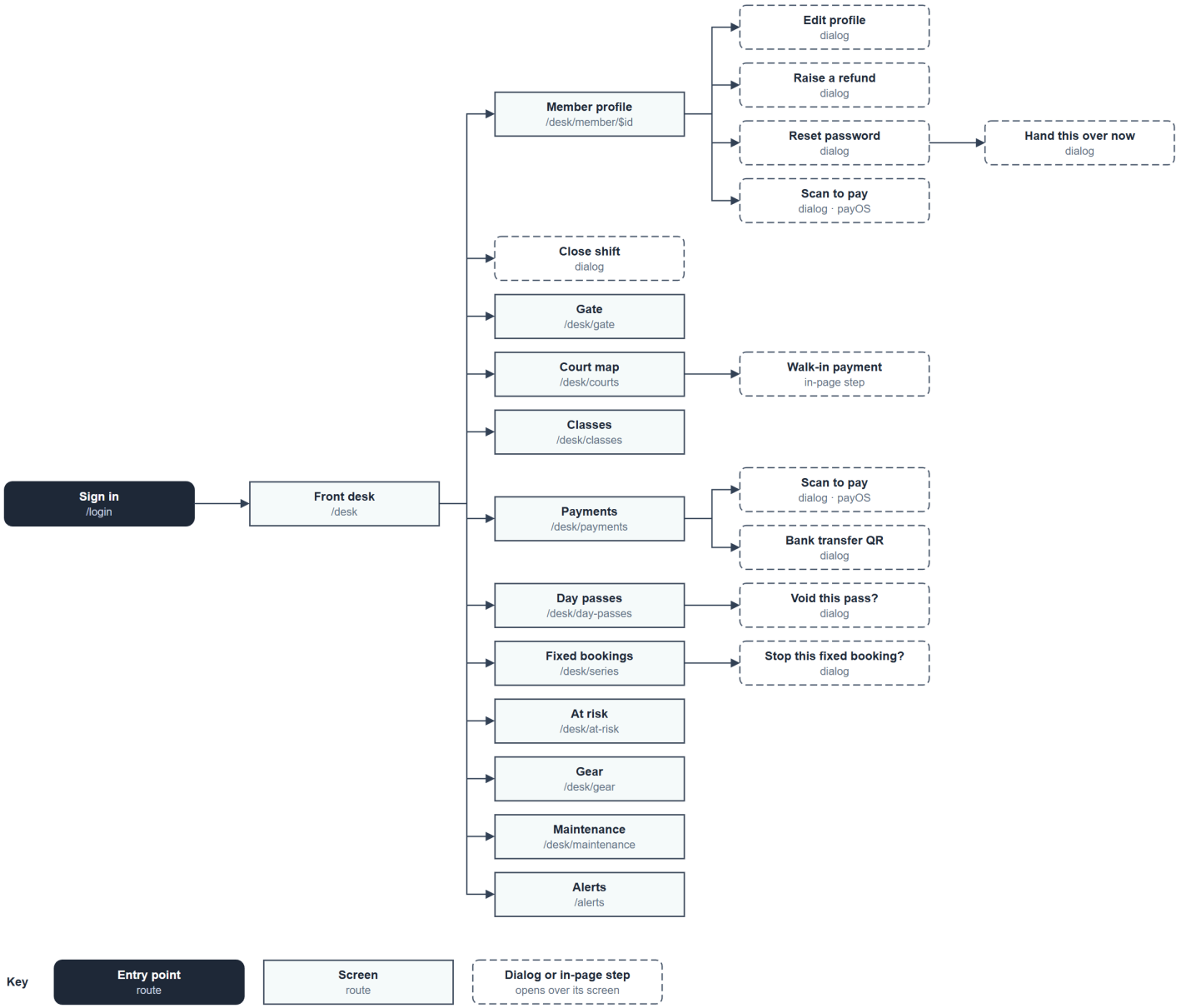


Figure 17. *Screens flow: front desk (receptionist)*

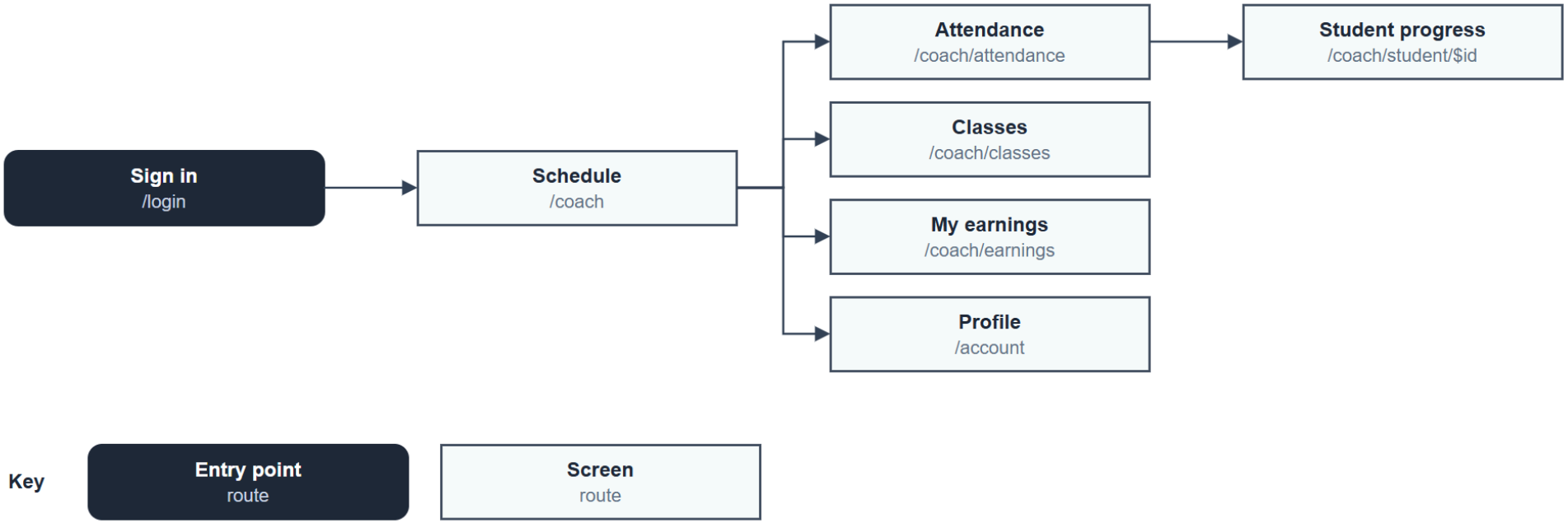


Figure 18. *Screens flow: coach portal*

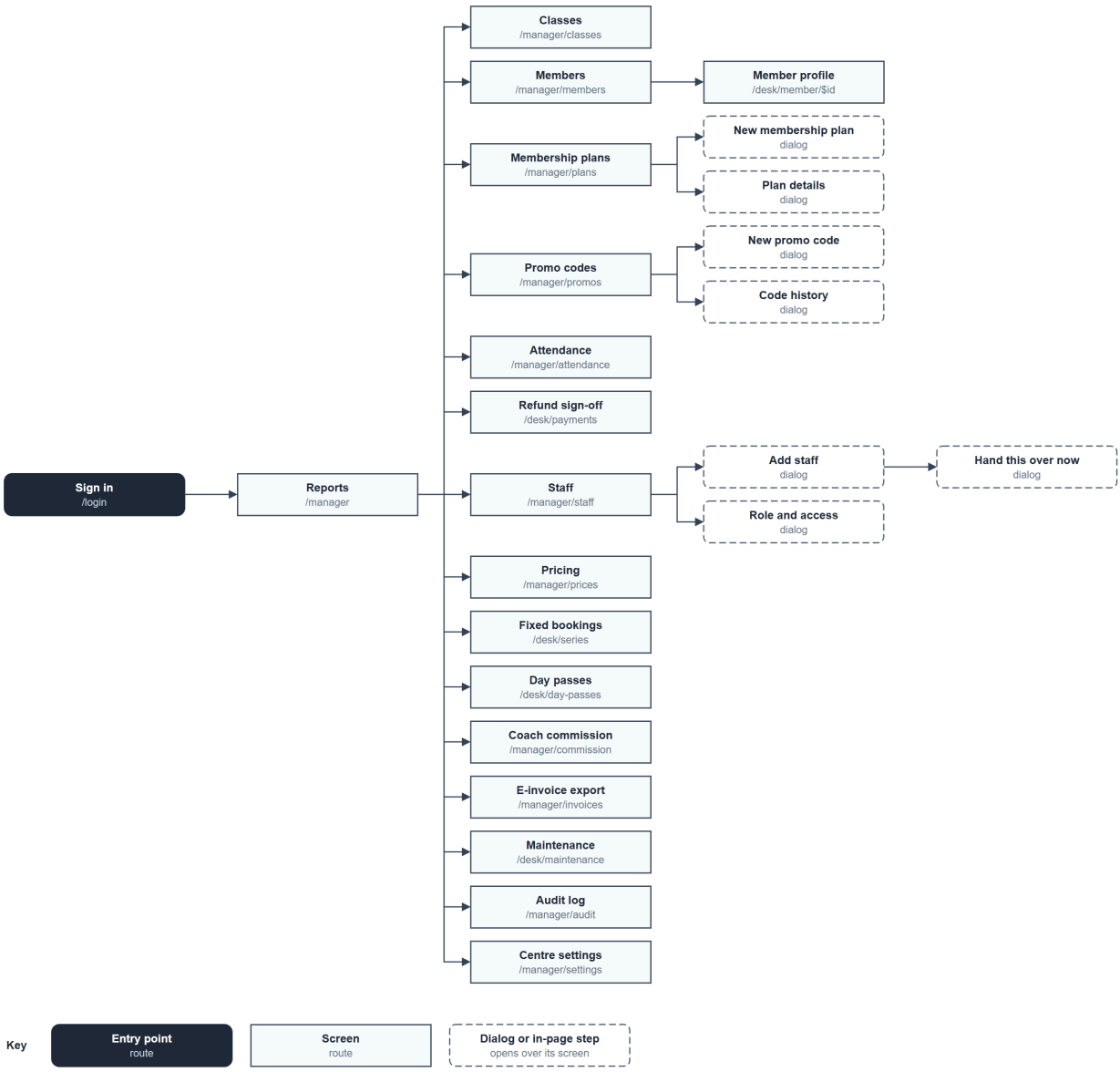


Figure 19. *Screens flow: manager portal*

1.2 Screen descriptions

The app has 44 routed screens. Each row names the route exactly as the app serves it.

#	Feature	Screen	Route	Description
1	Public	Landing page	`/`	Centre introduction, sports, prices, open classes and coach profiles.
2	Account	Sign in	`/login`	Sign in by phone or email and password. Each role lands on its own portal.
3	Account	Create an account	`/register`	Member sign-up with consent, confirmed by a one-time code.
4	Account	Forgot password	`/forgot`	Sends a reset code and sets a new password.
5	Account	Account settings	`/account`	Personal details, password, receipts and support requests.
6	Account	Alerts	`/alerts`	Staff alerts, such as an absence streak.
7	Payment	Payment result	`/pay/return`	Return page after payOS. Asks Arena3 to verify the payment, then shows the outcome.
8	Member	My schedule	`/app`	Upcoming bookings and classes, plan balance and waitlist offers.
9	Booking	Book a court	`/app/book`	Pick a date and sport, tap a free slot, hold it for 5 minutes and pay.
10	Classes	Classes	`/app/classes`	Enrol by sport. A full class offers a first-come waitlist.
11	Training	My progress	`/app/train`	Attendance, results, reviews, training plans and homework.
12	Plans	Membership plans	`/app/plans`	Order a plan here and pay at the desk.
13	Loyalty	My points	`/app/points`	Points balance and history.
14	Gate	Gate pass	`/app/pass`	QR pass whose code changes every minute.
15	Assistant	AI assistant	`/app/assistant`	Gemini answers grounded in the timetable, plans and coaches.
16	Notifications	Notifications	`/app/notifications`	In-app notices for the member.
17	Desk	Front desk	`/desk`	Find a member, sell a plan, take a payment, open and close the shift.
18	Desk	Member profile	`/desk/member/$id`	Plans, bookings and payments. Edit, refund, reset password, take a payOS payment.
19	Gate	Gate	`/desk/gate`	Scan a member's QR pass, or look the member up by phone.
20	Courts	Court map	`/desk/courts`	Live court grid. Sells a walk-in slot.
21	Classes	Classes (desk)	`/desk/classes`	Every class with its sessions, court, coach and roster.
22	Payments	Payments	`/desk/payments`	Plans awaiting payment, transfers to reconcile, refunds awaiting a manager, receipts.
23	Day passes	Day passes	`/desk/day-passes`	Sell, mark as used or void a day pass.
24	Bookings	Fixed bookings	`/desk/series`	Fixed weekly bookings over 4–12 weeks at 10% off.
25	Care	At risk	`/desk/at-risk`	Members whose attendance has dropped, with a contact log.

#	Feature	Screen	Route	Description
26	Gear	Gear	`/desk/gear`	Lend and take back equipment against stock.
27	Maintenance	Maintenance	`/desk/maintenance`	Work orders. An open order can block a court.
28	Coach	Schedule	`/coach`	The coach's own sessions by day, with court, class code and enrolment.
29	Coach	Attendance	`/coach/attendance`	Session register: Present, Late, Absent or Excused, plus results.
30	Coach	Student progress	`/coach/student/$id`	Goal, level, attendance totals, reviews, notes and homework.
31	Coach	Classes (coach)	`/coach/classes`	Classes the coach leads or assists, with plans and homework.
32	Coach	My earnings	`/coach/earnings`	Fee per class plus a fee per student present, by month.
33	Reports	Reports	`/manager`	Revenue by day, method, source and shift; capacity and members reports; export.
34	Classes	Classes (manager)	`/manager/classes`	Create, publish, edit and cancel classes. Clashes are blocked.
35	Members	Members	`/manager/members`	Every member with filters, plan and class count.
36	Plans	Membership plans	`/manager/plans`	Create, edit and withdraw plans. A plan is never deleted.
37	Promotions	Promo codes	`/manager/promos`	Create codes and read their redemption history.
38	Attendance	Attendance report	`/manager/attendance`	Attendance across classes, sessions and members.
39	Staff	Staff	`/manager/staff`	Issue, lock and re-role staff accounts; reset passwords.
40	Pricing	Pricing	`/manager/prices`	Price rules by sport, court, day type and hour band. A new price applies only to new sales.
41	Commission	Coach commission	`/manager/commission`	Close a month's coach commission and mark it paid.
42	Invoices	E-invoice export	`/manager/invoices`	Export invoices as a spreadsheet or XML.
43	Audit	Audit log	`/manager/audit`	Audit entries filtered by actor, action, entity and date.
44	Settings	Centre settings	`/manager/settings`	Opening hours, slot length, windows, limits, VAT, rounding and feature flags.

1.3 Screen authorization

Each portal layout checks the signed-in role before it renders. A wrong role is sent to its own home; no session is sent to Sign in.

Screens	Guest	Member	Receptionist	Coach	Manager
Landing page, Sign in, Create an account, Forgot password, Payment result	X	X	X	X	X
Account settings		X	X	X	X
Alerts			X	X	X
Member portal (`/app/*`, screens 8–16)		X			
Front desk (`/desk/*`, screens 17–27)			X		X

Screens	Guest	Member	Receptionist	Coach	Manager
Coach portal (`/coach/*`, screens 28–32)				X	X
Manager portal (`/manager/*`, screens 33–44)					X

The API checks every request again on the server, so hiding a button is never the only control. These actions are narrower than the screen that shows them:

Action	Allowed roles	Refused with
Approve or reject a refund above the limit	Manager	403 FORBIDDEN
Mark a bank transfer as received or rejected	Receptionist, Manager	403 FORBIDDEN
Raise a payOS link for a plan order	Receptionist, Manager	403 FORBIDDEN
Raise a payOS link for a held court	Member (own hold only), Receptionist, Manager	403 FORBIDDEN
Write a register for a session	Coach (own classes only), Manager	403 FORBIDDEN
Correct a closed register	Manager, with a reason	422 BR-53
Read another member's bookings or receipts	Receptionist, Manager	403 FORBIDDEN

1.4 Non-screen functions

Ten background jobs keep holds, bookings, plans and messages correct without anyone pressing a button. A single probe query checks every 15 seconds whether any job has work.

#	Function	Trigger	Description
1	Expire holds	Every 15 s, when due	Cancels court holds past their hold time and frees the court.
2	Mark no-shows	Every 15 s, when due	Marks a confirmed booking as a no-show 10 minutes after its start without check-in.
3	Complete bookings	Every 15 s, when due	Closes an in-use booking 10 minutes after it ends.
4	Send notifications	Every 15 s, when due	Sends up to 50 queued messages. A message is marked sent only on delivery and stops after 5 tries.
5	Withdraw payment links	Every 15 s, when due	Cancels a payOS link whose court hold has lapsed. Runs only when payOS is configured.
6	Expire waitlist offers	Every 15 s, when due	Expires an unanswered offer and offers the seat to the next member.
7	Lock attendance	Every 15 s, when due	Closes a session's register 2 hours after it ends.
8	Expire plans	Every 15 s, when due	Marks a plan expired the day after its end date.
9	Plan expiry reminders	Every 15 s, when due	Sends one in-app reminder 7 days, 3 days and 0 days before a plan ends.
10	Generate sessions	Every 10 minutes	Creates upcoming sessions for every open class.
11	payOS webhook	Called by payOS	Verifies the signature and amount, then posts the payment through the normal settle path.

The job loop starts with the first API request on each server instance. Each job with work runs in its own transaction, so one failure cannot roll back another.

1.5 Entity relationship diagram

The conceptual ERD shows 17 core entities in Chen notation, with a verb on each relationship and 1, N or M cardinality.

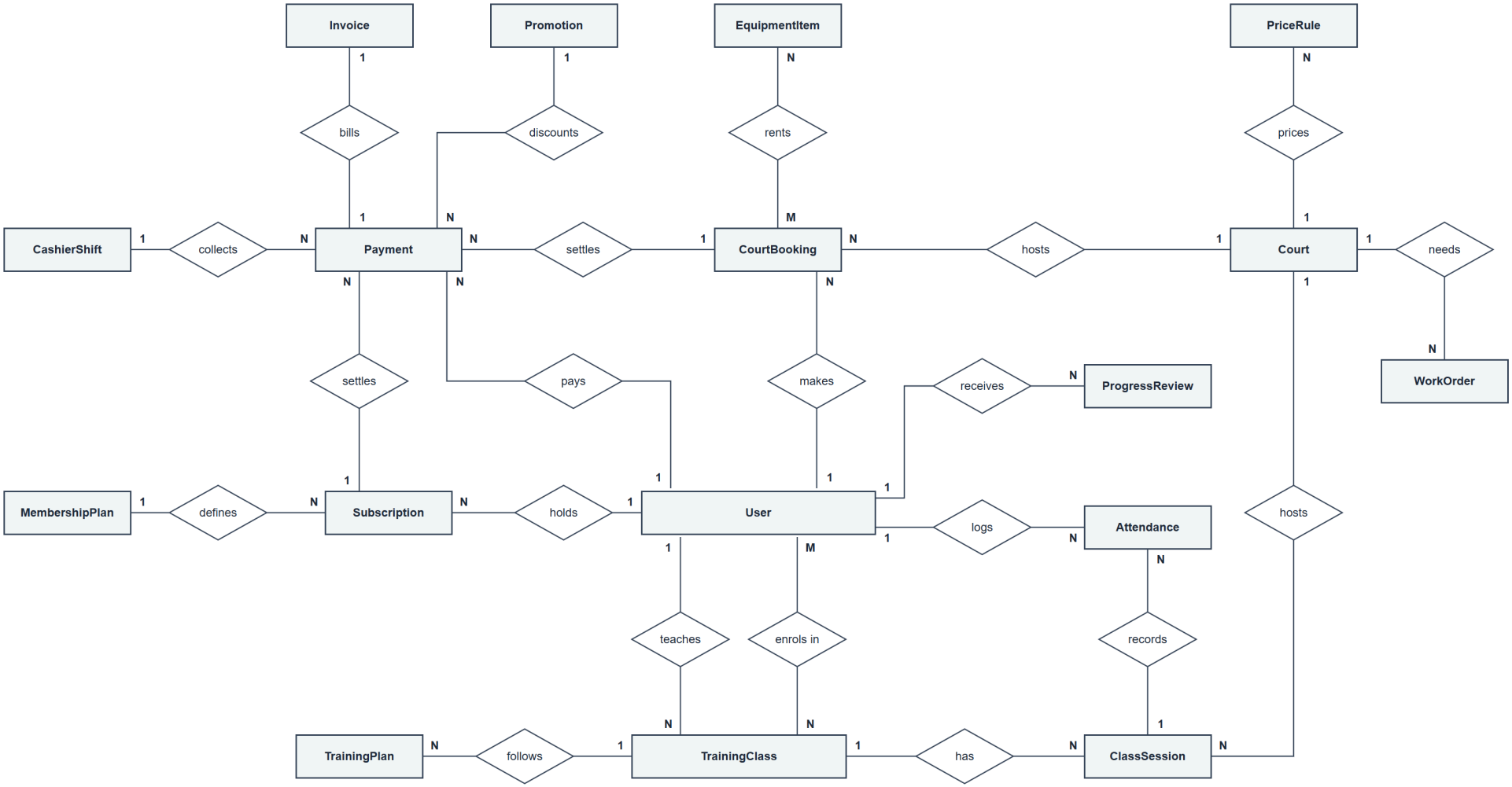


Figure 20. *Conceptual entity relationship diagram*

#	Entity	Table	Description
1	User	`users`	Every person with an account, with one role: member, receptionist, coach or manager.
2	Court	`courts`	A playable court for one sport, with a status of ready, maintenance or closed.
3	PriceRule	`price_rules`	Price per hour by sport, court class, day type and hour band.
4	CourtBooking	`court_bookings`	A court-time owned by a member or a walk-in, from hold to completed.
5	Payment	`payments`	Money in or out, with method, amount, VAT, shift and capture mode.
6	Invoice	`invoices`	The receipt issued for a payment, with a gapless code.
7	Promotion	`promotions`	A promo code with its discount, validity and usage limits.
8	CashierShift	`cashier_shifts`	One receptionist's till session, from opening float to counted close.
9	MembershipPlan	`membership_plans`	A plan on sale: price, duration, court hours or sessions, sport scope.
10	Subscription	`subscriptions`	A member's copy of a plan, with start and end dates and balances.
11	TrainingClass	`classes`	A coached class with a coach, a court, a weekly timetable and a capacity.
12	ClassSession	`sessions`	One dated meeting of a class.
13	Attendance	`attendance`	One member's mark for one session.
14	TrainingPlan	`training_plans`	Up to 12 timed blocks for a class, a session or one student.
15	ProgressReview	`progress_reviews`	An append-only review scoring technique, fitness and attitude.
16	EquipmentItem	`equipment_items`	Rentable gear with stock on hand.
17	WorkOrder	`work_orders`	A maintenance job that can block a court while it is open.

1.6 Class diagrams

Four class diagrams refine the ERD by subsystem, with attributes, operations, enumerations and multiplicities.

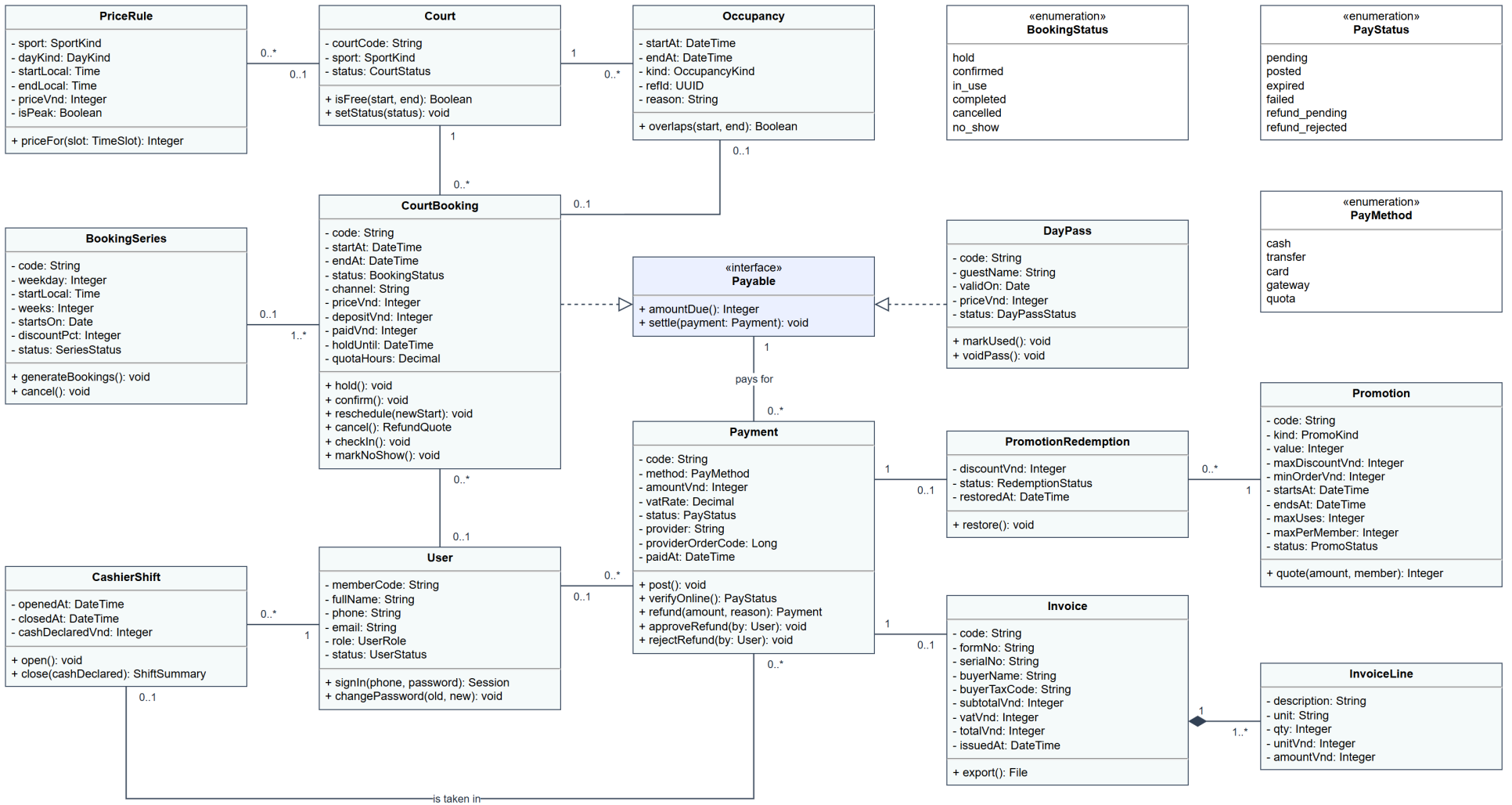


Figure 21. *Class diagram: booking and payment*

Figure 21 covers Court, PriceRule, Occupancy, CourtBooking, BookingSeries, DayPass, Payment, CashierShift, Promotion, PromotionRedemption, Invoice and InvoiceLine. CourtBooking, DayPass and Subscription realise the Payable interface.

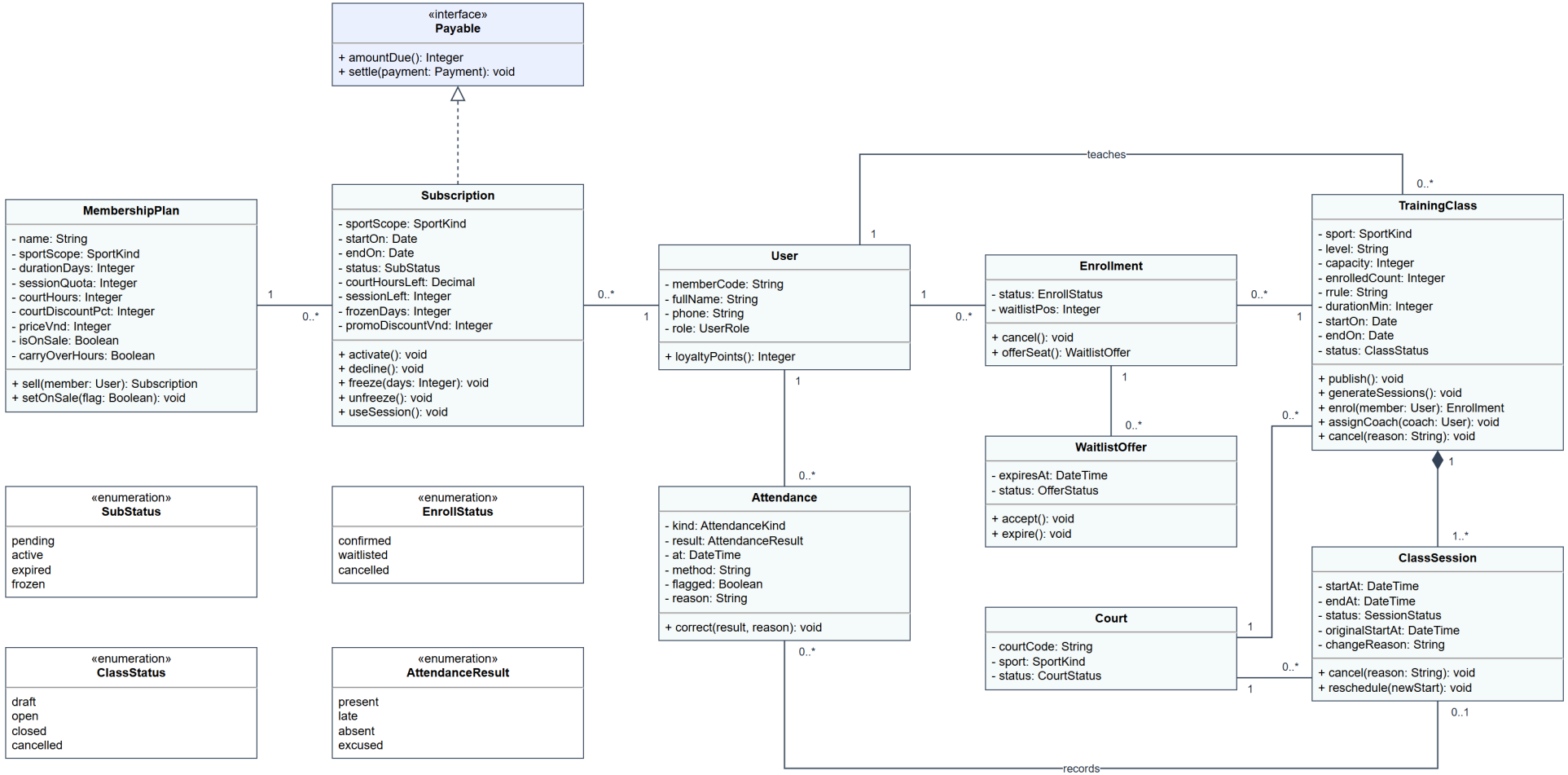


Figure 22. *Class diagram: membership and classes*

Figure 22 covers MembershipPlan, Subscription, TrainingClass, ClassSession, Enrollment, WaitlistOffer and Attendance, with their status enumerations.

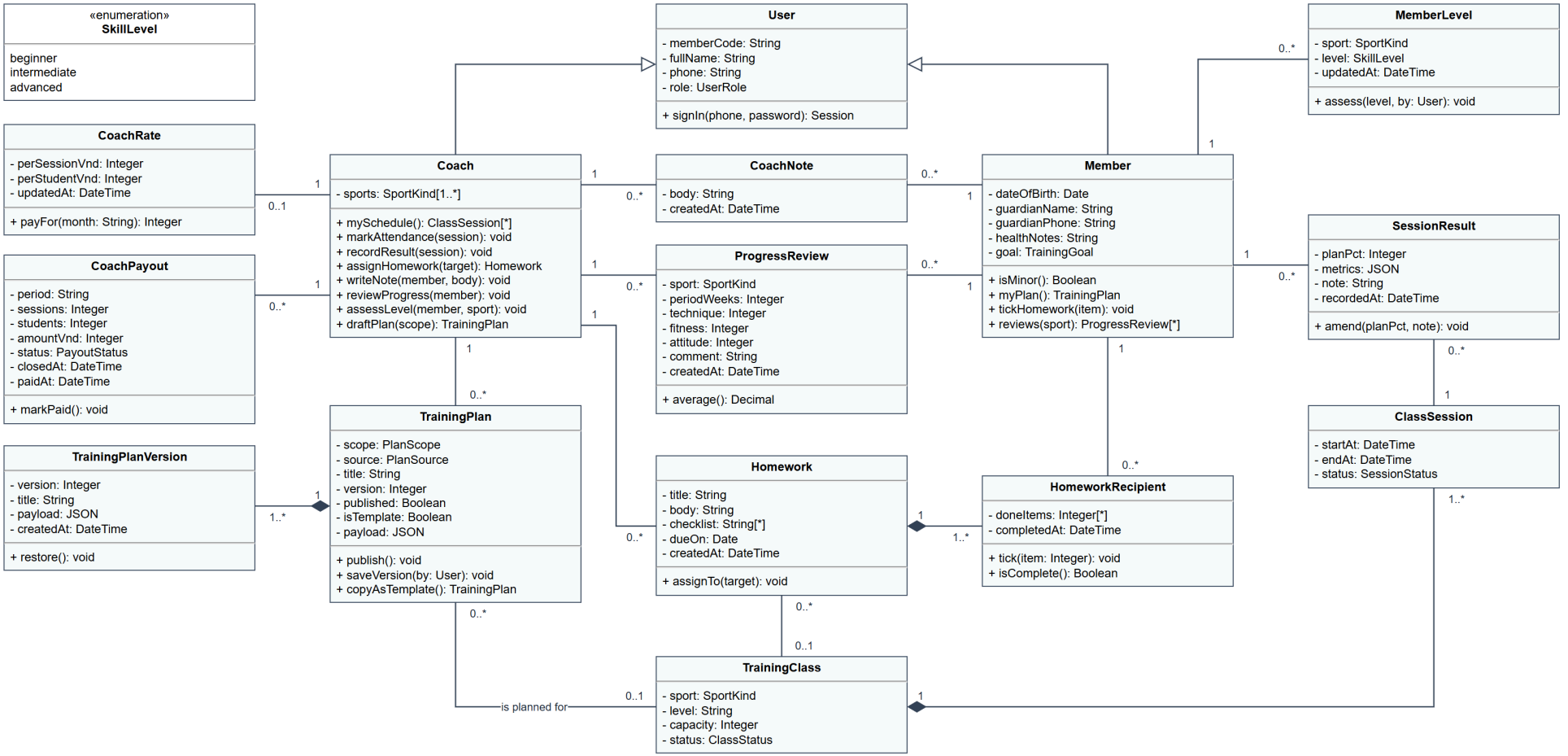


Figure 23. *Class diagram: coaching and training*

Figure 23 specialises User into Coach and Member. It covers CoachRate, CoachPayout, MemberLevel, SessionResult, TrainingPlan, TrainingPlanVersion, Homework, HomeworkRecipient, ProgressReview and CoachNote.

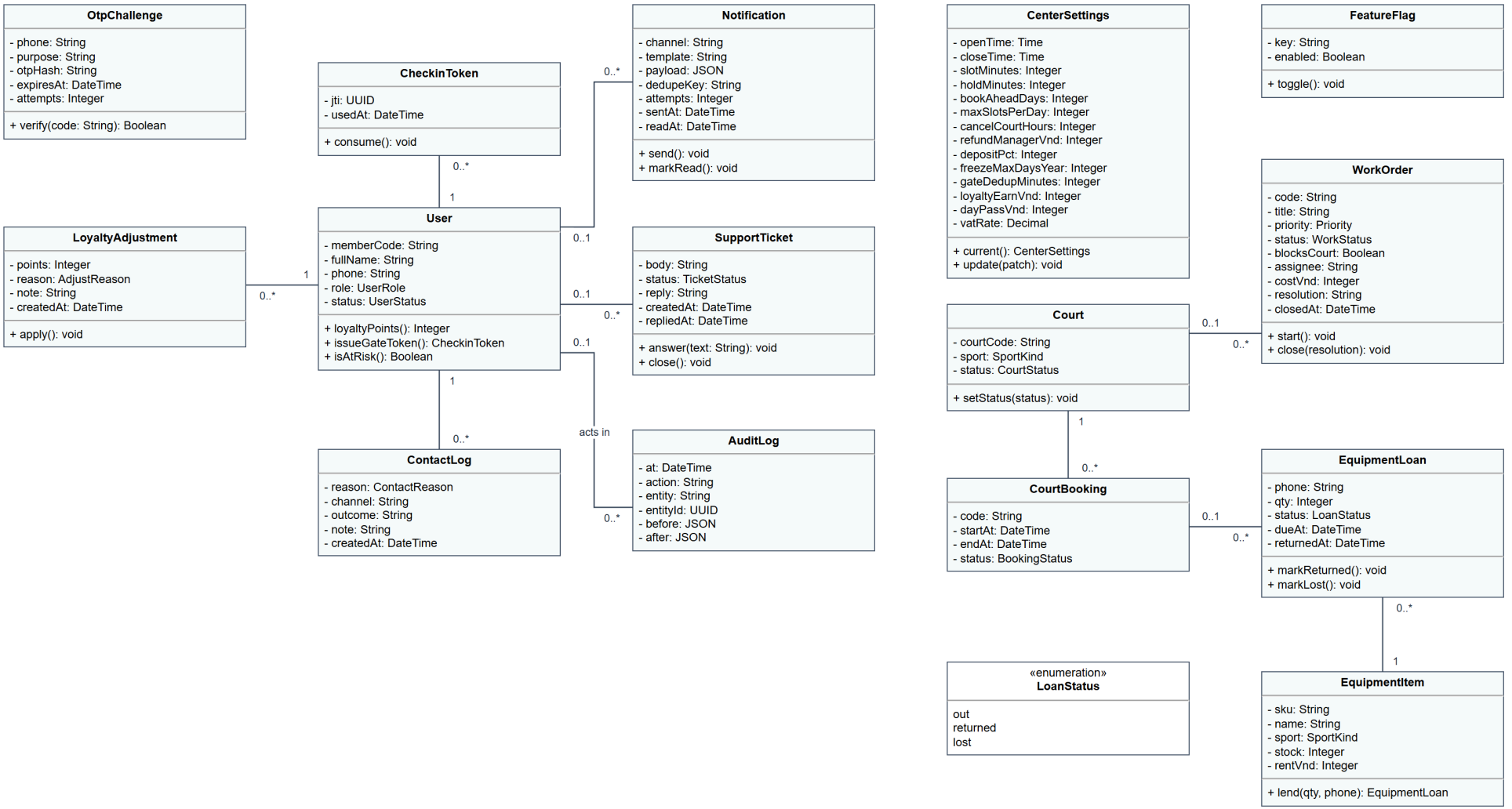


Figure 24. *Class diagram: centre operations*

Figure 24 covers CenterSettings, FeatureFlag, OtpChallenge, CheckinToken, Notification, SupportTicket, ContactLog, LoyaltyAdjustment, AuditLog, WorkOrder, EquipmentItem and EquipmentLoan.

2. Use case specifications

This section gives full specifications for the five use cases that involve identity, money or access. Each follows the FPT template and links to its sequence diagram or activity diagram. The remaining use cases follow the screen descriptions in 1.2 and the business rules in V.1.

Every refusal named below reaches the screen as a sentence, with the error code and, where one applies, the `BR-nn` rule that refused it.

2.1 UC-02 Register a member account

Field	Detail
UC ID and name	UC-02 Register a member account (includes UC-09 Verify a one-time code)
Created by / date	Arena3 team / 2026-10-11
Primary actor	Guest
Secondary actors	Email service (SMTP)
Trigger	The guest selects Create an account on `/register`.
Description	A guest creates a member account and proves the email address with a one-time code.
Preconditions	The guest is signed out. The phone and email have no account yet.
Postconditions	A member account exists, the guest is signed in for seven days, and an audit row is written.
Priority	Must have
Frequency of use	Once per new member
Business rules	BR-01, BR-02, BR-07, BR-08

Normal flow

1. The guest enters full name, phone, email, password and date of birth.
2. The guest accepts the terms and the data-privacy notice, then submits.
3. The system checks every field and confirms the phone and email are unused.
4. The system stores a sign-up challenge for five minutes and emails a one-time code.
5. The guest enters the code on the same page (UC-09).
6. The system checks the code, creates the member account and returns a session token.
7. The system opens My schedule.

Alternative flows

- 1a. The guest is under 16: the guest also enters a guardian's name and phone (BR-07).

Exceptions

- 3a. A field is malformed: `VALIDATION`, and the field to correct is marked.

- 3b. The phone or email already has an account: refused with BR-01.
- 3c. The password is shorter than 8 characters or lacks letters and digits: BR-02.
- 3d. The terms are not accepted: BR-08.
- 6a. The code is wrong: the attempt is counted and the guest may try again.
- 6b. The code is older than five minutes: the guest submits the sign-up form again to receive a new code.
- 6c. Five wrong attempts: the challenge is locked for 15 minutes (`RATE_LIMITED`).

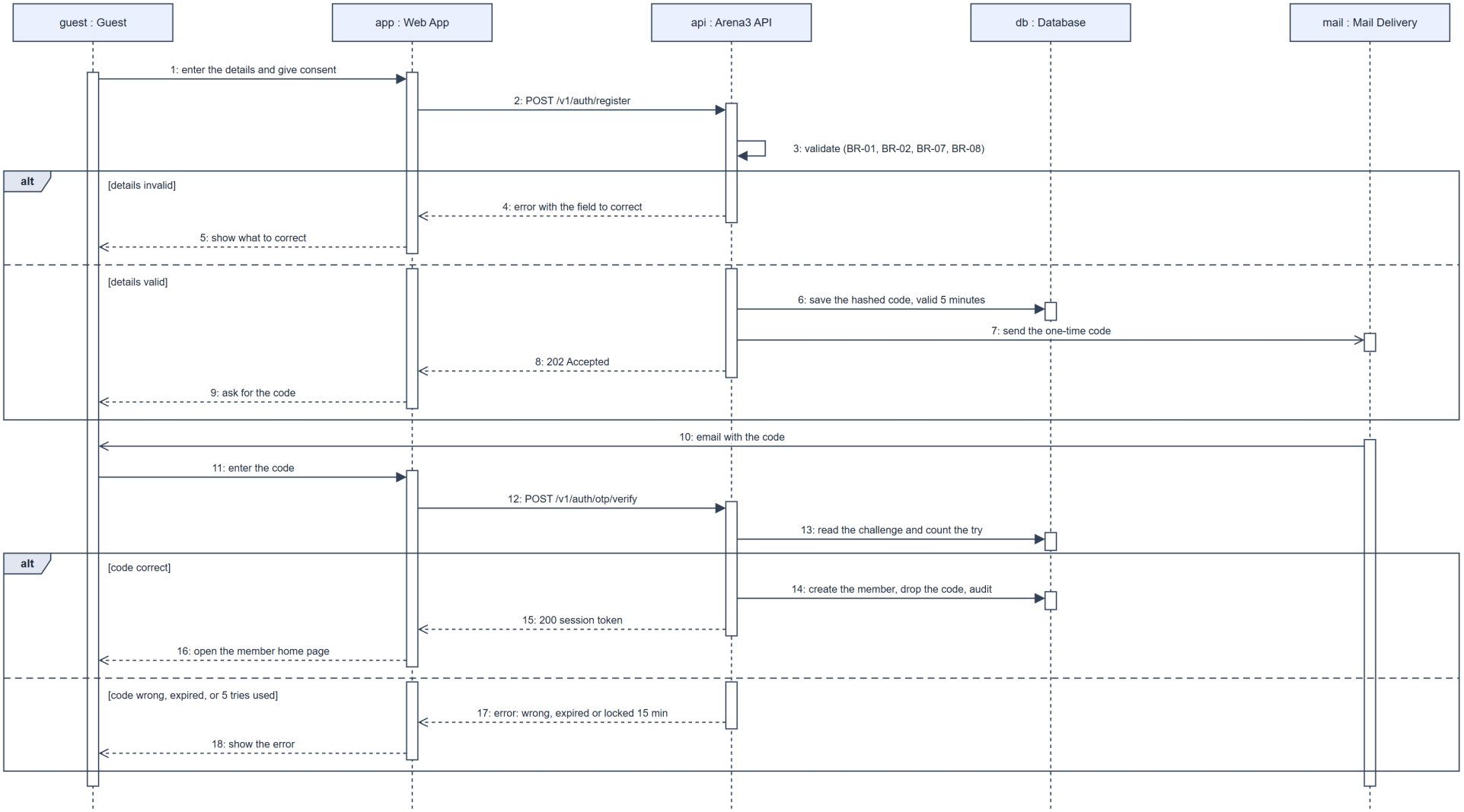


Figure 25. *Sequence diagram: register a member account*

2.2 UC-12 Book a court and pay online

Field	Detail
UC ID and name	UC-12 Book a court (includes UC-20 Pay online by payOS QR; extended by UC-21 and UC-22)
Created by / date	Arena3 team / 2026-10-11
Primary actor	Member
Secondary actors	payOS
Trigger	The member selects a free slot on Book a court (`/app/book`).
Description	The member holds a court slot for five minutes and pays the amount due by VietQR to confirm it.
Preconditions	The member is signed in. The slot is in the future, within seven days and inside opening hours. The court is ready.
Postconditions	Success: the booking is confirmed and the payment is posted with a receipt code. Failure: the hold lapses and the slot is free again.
Priority	Must have
Frequency of use	Many times a day
Business rules	BR-32, BR-34B, BR-35, BR-39C, BR-39H, BR-44, BR-45, BR-46, BR-66

Normal flow

1. The member chooses a sport and a date, then a free slot on the court grid.
2. The system checks the booking window, the court status and the daily cap of two slots.
3. The system prices the slot from the price rules and holds it for five minutes.
4. The member may apply a promo code (UC-22).
5. The member chooses Pay online. The system creates a payOS link and shows the VietQR in the Scan to pay dialog.
6. The member scans the QR with a banking app and pays.
7. payOS confirms by a signed webhook, or the dialog's check every three seconds reads the link as paid.
8. The system compares the amount with the amount due, posts the payment and confirms the booking.
9. The dialog shows the receipt, and the booking appears in My schedule.

Alternative flows

- 2a. The slot clashes with one of the member's classes: the system asks for confirmation (BR-39C). The member may hold it anyway.
- 3a. The slot is at peak time: the member pays a 30% deposit now, and the desk collects the balance on arrival.
- 3b. The member already holds a slot: the new hold is checked in full before it replaces the old one (BR-39H).

- 5a. The member pays by bank transfer instead (UC-21). The booking is confirmed only after the desk reconciles the transfer.
- 5b. The member selects Open the payment page instead and pays on the payOS page.

Exceptions

- 2b. The slot is in the past (BR-66), more than seven days ahead (BR-32), or outside opening hours (BR-35).
- 2c. The court is not ready (BR-35), or the member already holds two slots that day (BR-32).
- 3c. No price rule matches the slot: refused with BR-34B; the system never invents a price.
- 3d. Another member took the slot first: `409 CONFLICT_SLOT`.
- 4a. The promo code is unknown, expired or out of scope (BR-44), or already used up (BR-45).
- 4b. A code that does not stack meets a plan discount: the better of the two applies (BR-46).
- 8a. The hold lapsed before payment was confirmed: `409 HOLD_EXPIRED`, and the member books again.

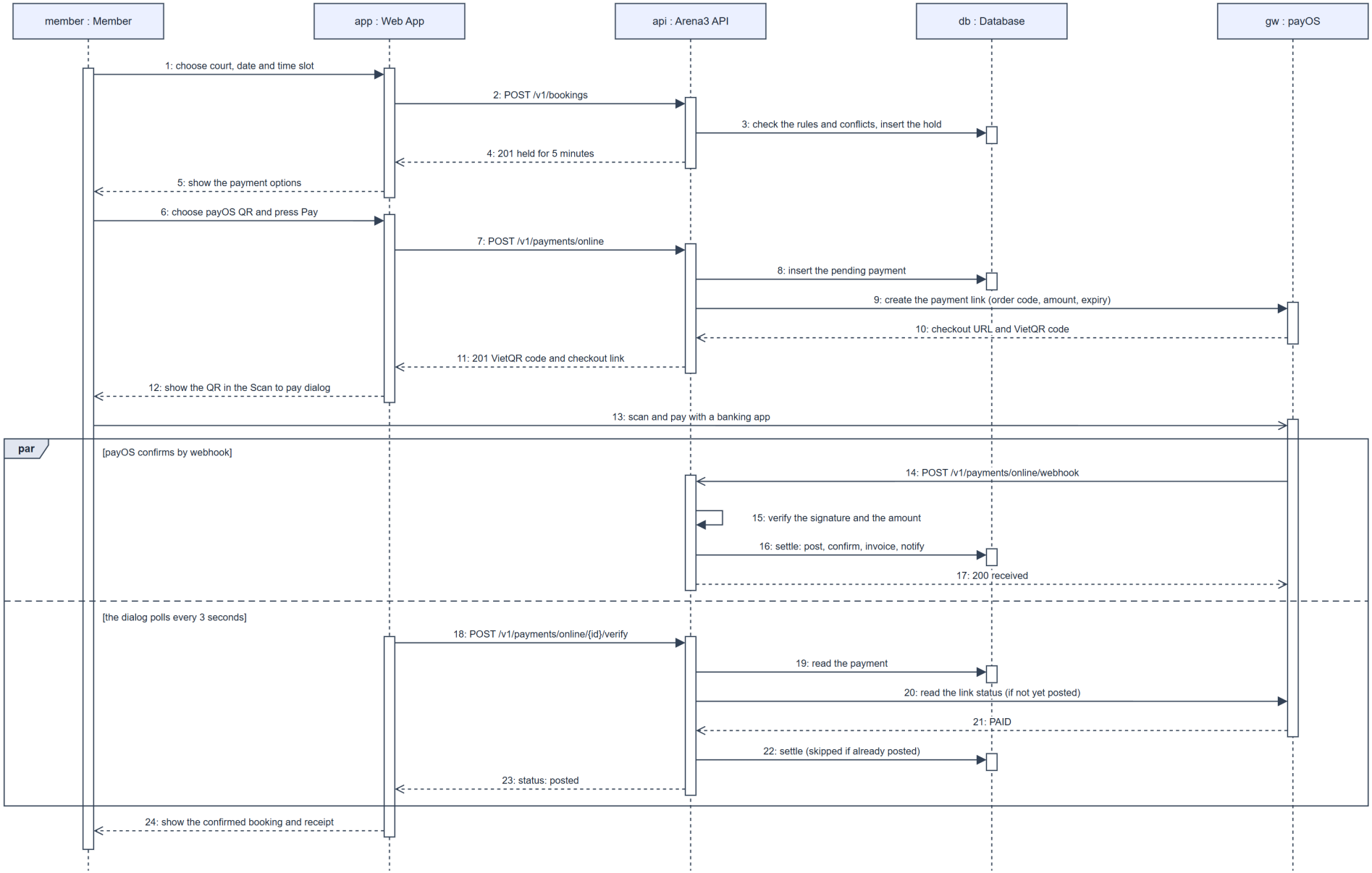


Figure 26. *Sequence diagram: book a court and pay by payOS QR*

2.3 UC-24 Enrol in a class

Field	Detail
UC ID and name	UC-24 Enrol in a class (extended by UC-33 Join the class waitlist and UC-25 Accept a waitlist offer)
Created by / date	Arena3 team / 2026-10-11
Primary actor	Member
Secondary actors	Receptionist or manager, enrolling on the member's behalf
Trigger	The member selects Enrol on a class in Classes (`/app/classes`).
Description	The member takes a seat in an open class, or joins its waitlist when the class is full.
Preconditions	The member is signed in and holds an active plan covering the class sport.
Postconditions	The member is enrolled and the sessions appear in My schedule, or the member is waitlisted with a position.
Priority	Must have
Frequency of use	Daily
Business rules	BR-12, BR-18, BR-22, BR-24, BR-25, BR-67

Normal flow

1. The member opens Classes and chooses an open class.
2. The system checks that a plan covers the sport and has sessions left.
3. The system checks the class does not clash with the member's other classes.
4. A seat is free: the system enrols the member and adds the class sessions to My schedule.

Alternative flows

- 4a. The class is full: the system adds the member to the waitlist (UC-33) and shows the position.
- 4b. A seat frees up: the next member in line receives an offer valid for two hours.
- 4c. The member accepts the offer (UC-25): the system enrols the member if the seat is still free.

Exceptions

- 1a. The class is not open for enrolment: BR-67.
- 2a. No active plan covers the sport: BR-12.
- 2b. The plan has no sessions left: BR-18.
- 3a. The class clashes with another class, or the member is already enrolled: BR-24.
- 4d. The offer expired before acceptance: BR-25. The seat passes to the next member in line.

The activity diagram in Figure 4 (I.3.3) shows this flow with its waitlist branch.

2.4 UC-46 Check a member in at the gate

Field	Detail
UC ID and name	UC-46 Check a member in at the gate (includes UC-55 Scan QR; extended by UC-56 Look up by phone)
Created by / date	Arena3 team / 2026-10-11
Primary actor	Receptionist
Secondary actors	Member, who shows the gate pass
Trigger	The member shows the gate pass QR at the front desk.
Description	The receptionist admits a member and the system records the visit, flagging any override.
Preconditions	The receptionist is signed in on Gate (`/desk/gate`). The member opens Gate pass (`/app/pass`).
Postconditions	One visit row exists for the member within the 120-minute window. Any override is flagged and audited.
Priority	Must have
Frequency of use	Hundreds of times a day
Business rules	BR-54, BR-71, BR-72, BR-73

Normal flow

1. The member opens Gate pass, which shows a short-lived QR code.
2. The receptionist scans the code on the Gate screen (UC-55).
3. The system checks the code is a valid member code that has not been used.
4. The system checks today's entitlement: a usable plan, a court booking or a class session today.
5. The system records the check-in time, returns any warnings and shows a green pass.

Alternative flows

- 2a. The code cannot be scanned: the receptionist looks the member up by phone (UC-56). The receptionist picks why: no phone, dead battery, QR failed, forgot, or other.
- 5a. The member is not entitled: the system lists the reasons and asks for an override.
- 5b. The receptionist picks a reason: expired but allowed, manager approval, guest pass, renewing, or other. The visit is flagged and audited, and the screen shows an amber pass (BR-72).
- 5c. The member already checked in within 120 minutes: the system returns that visit instead of writing a second (BR-72).

Exceptions

- 3a. The code is invalid or already used: BR-71, and the member refreshes the code.
- 3b. The desk's own self check-in code is scanned at the desk: BR-71.

Warnings cover a frozen plan, an expired plan, a plan ending within seven days and a plan with no sessions left. Walking through the gate never spends a session (BR-73) and never fills a class register (BR-54).

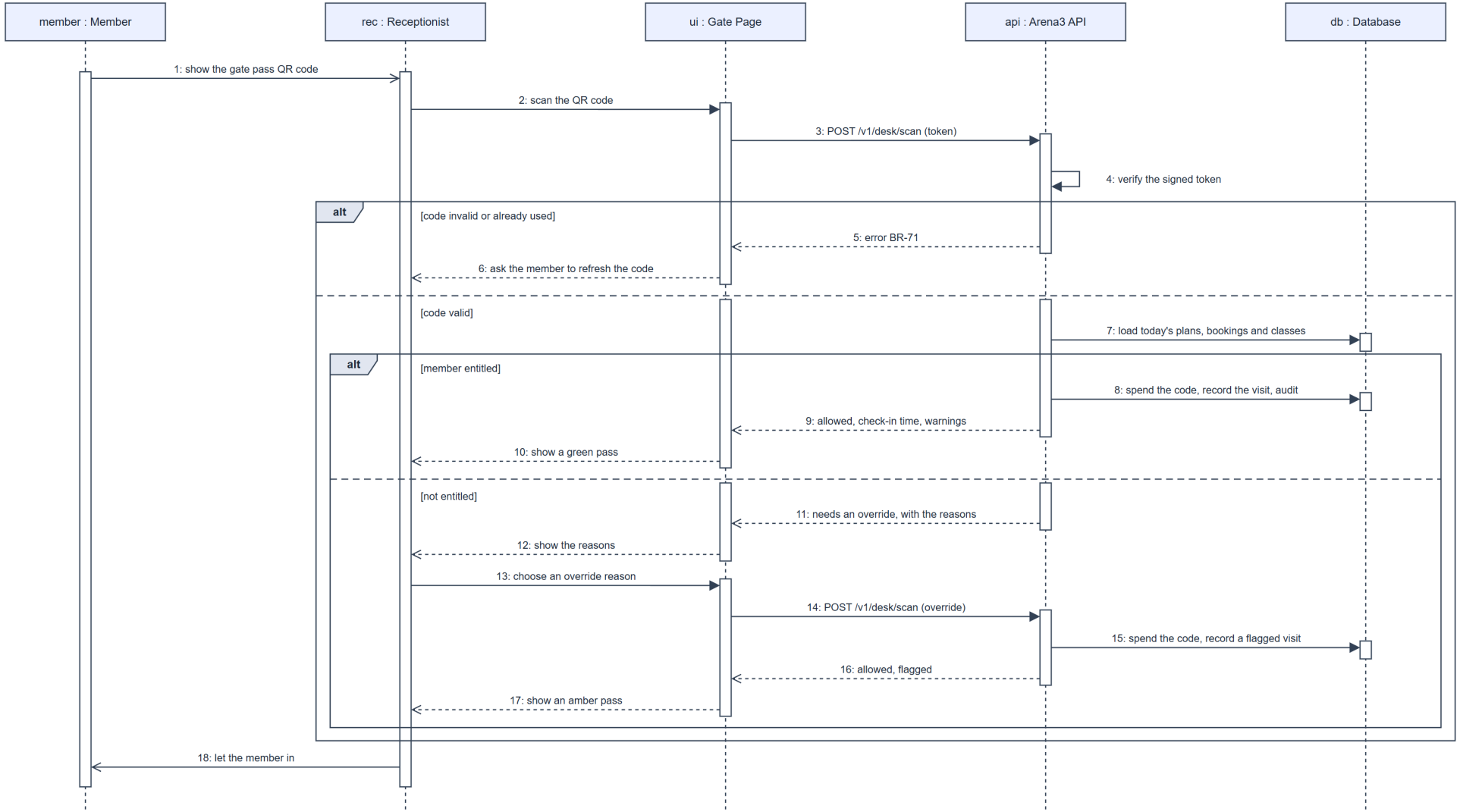


Figure 27. *Sequence diagram: check a member in at the gate*

2.5 UC-43 Raise a refund

Field	Detail
UC ID and name	UC-43 Raise a refund (extended by UC-80 Approve or reject a refund)
Created by / date	Arena3 team / 2026-10-11
Primary actor	Receptionist
Secondary actors	Manager, who signs off large refunds; Member, who is notified
Trigger	The receptionist chooses Raise a refund on a payment in Member profile.
Description	Staff give back all or part of a posted payment, and large refunds wait for a second person.
Preconditions	The receptionist is signed in. The payment is posted and has money left to refund.
Postconditions	A negative payment row exists with its own receipt code, either posted or pending approval. An audit row is written.
Priority	Must have
Frequency of use	A few times a week
Business rules	Refund limit `refund_manager_vnd` (1,000,000đ); separation of duties

Normal flow

1. The receptionist opens the member profile, selects a posted payment and chooses Raise a refund.
2. The receptionist enters the amount and a reason.
3. The system works out what is still refundable: the payment, less refunds already posted or pending.
4. The amount is under 1,000,000đ: the system posts the refund with its own receipt code.
5. The system writes an audit row and shows the refund receipt.
6. If the order is now refunded in full, a used promo code is given back to the member.

Alternative flows

- 4a. The amount is 1,000,000đ or more: the refund is saved as pending approval.
- 4b. A manager opens Refund sign-off (`/desk/payments`) and approves it (UC-80). The system posts it and notifies the member.
- 4c. The manager rejects it with a reason. The system closes it and notifies the member.
- 4d. A manager raises the refund: it is posted directly, whatever the amount.

Exceptions

- 3a. The payment is already refunded in full, or the amount is larger than what remains: `VALIDATION`.
- 4e. The person who raised the refund tries to approve it: `FORBIDDEN`, and another manager must sign it off.
- 4f. Two managers decide the same refund at once: the second receives `CONFLICT_STATE`.

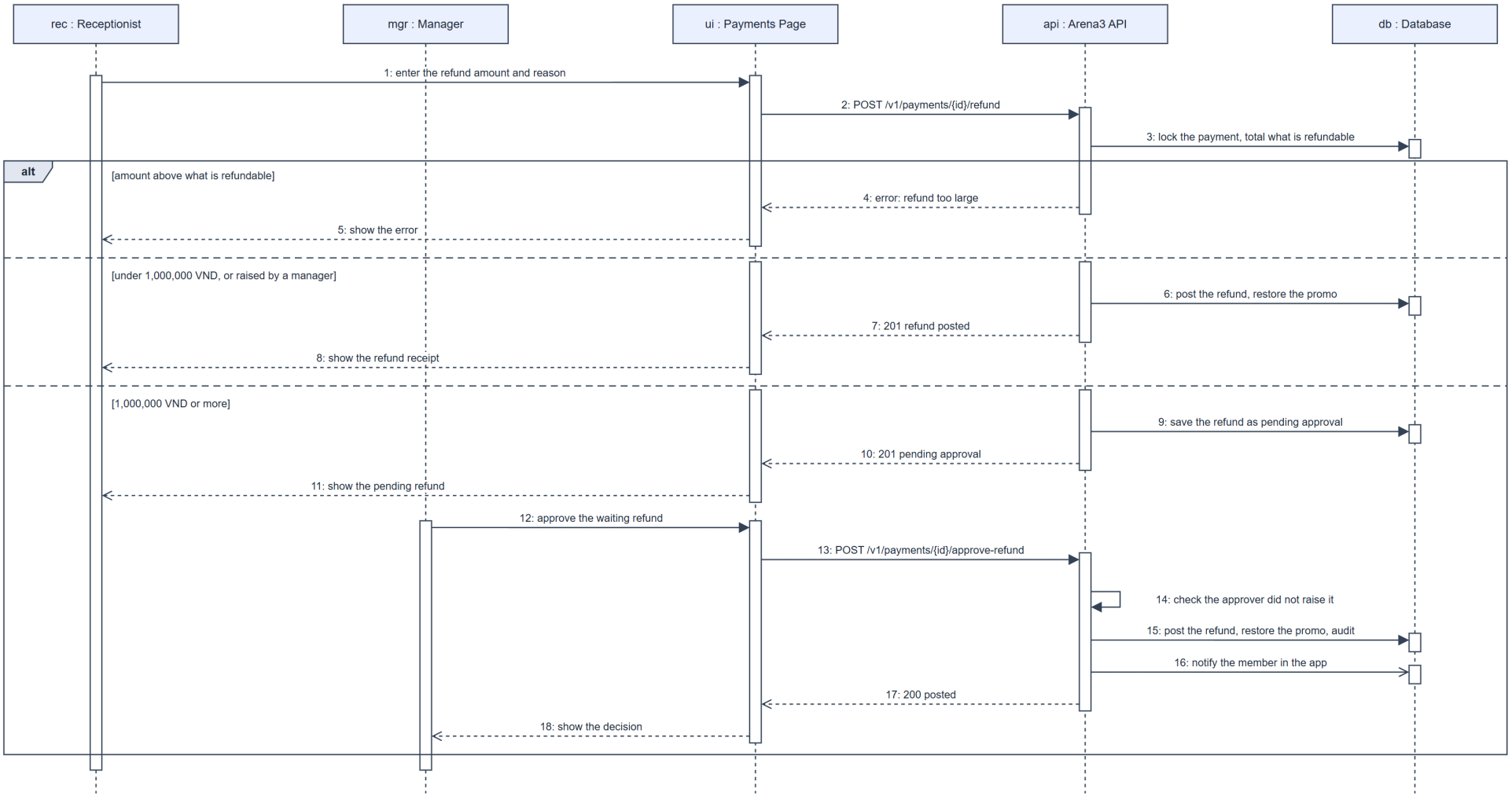


Figure 28. *Sequence diagram: raise and approve a refund*

IV. Non-functional requirements

Arena 3 must keep money and schedules correct under concurrency, respond quickly on a phone, and protect member data. This section names the external interfaces and the quality attributes that make this so.

1. External interfaces

1.1 User interfaces

Arena 3 is one responsive web application with four portals, and every screen works from a 375 px phone width.

- The interface is available in English and Vietnamese, and the user switches language in the header.
- Running text uses Be Vietnam Pro, which renders Vietnamese diacritics correctly. Headings use Newsreader and figures use Anton.
- Amounts are written in VND with a lowercase đ, for example 1,500,000đ.
- A refusal is shown as a sentence that says what to do instead. The error code travels with it for support.
- Motion is decorative only, and no action is discoverable only through animation.

1.2 Software interfaces

The application talks to five outside systems, and each has a defined fallback.

Interface	Purpose	Protocol and data	Failure behaviour
Arena 3 API (`/v1`)	Every screen reads and writes through it	JSON over HTTPS with a bearer token; served inside the web app	Errors return `{ code, message }`, plus `field` or `br` where one applies
payOS	Payment links, VietQR codes, link status and payment webhooks	HTTPS REST; the webhook is signed with the checksum key	A payment posts only when payOS confirms it, and the amount must equal the amount due
Google Gemini (`gemini-2.5-flash`)	Answers in the AI assistant	HTTPS; the prompt carries centre data only, never member data	12-second timeout, then xAI Grok, then rule-based answers
Email (SMTP through nodemailer)	One-time codes and notifications	SMTP with the centre's sender address	Codes are sent at once; notifications are queued and retried up to five times
PostgreSQL 16	All records	Neon in production; embedded PGLite when no `DATABASE_URL` is set	Schema changes are forward-only numbered migrations, each run in one transaction

The application is hosted on Vercel. Background jobs start with the first API request and run in a loop every 15 seconds.

1.3 API error codes

Every API error uses one of these codes, so a screen can react to the code rather than parse the text.

HTTP	Code	Meaning
400	`VALIDATION`	A value is malformed; `field` names the input to correct
401	`UNAUTHENTICATED`	No session, or the session has expired
403	`FORBIDDEN`	Wrong role, or another member's data. With `br: "BR-62"`, the module is switched off
404	`NOT_FOUND`	The record does not exist
409	`CONFLICT_SLOT`	Another person took that court time first
409	`HOLD_EXPIRED`	The five-minute hold ran out
409	`CONFLICT_STATE`	The record is in the wrong state for this action
422	`BR_VIOLATION`	A named business rule refused the request; `br` carries the rule code
422	`VALIDATION`	The request is well formed but not acceptable in the current state
429	`RATE_LIMITED`	Too many attempts in a short time

2. Quality attributes

2.1 Correctness and reliability

Two bookings can never own the same court time, and money never disagrees with the schedule.

ID	Requirement
NFR-01	No two bookings may own the same court time, under any concurrency. Both booking paths lock the court row, and a database trigger rejects any overlap.
NFR-02	No coach may be scheduled into two overlapping sessions.
NFR-03	A confirmed booking always has its payment and invoice, written in the same transaction.
NFR-08	Any request that takes money or changes a schedule accepts an `Idempotency-Key`. A replay returns the first result and never charges twice.
NFR-09	With no `DATABASE_URL`, the application migrates and seeds an embedded PGLite database, so a fresh copy runs offline.
NFR-10	Losing the assistant's model degrades only the assistant, which falls back to rule-based answers.

2.2 Performance

The screens staff and members open most often must cost as few database round trips as possible.

ID	Requirement
NFR-04	`GET /v1/me` issues its independent reads concurrently, so a sign-in costs one round trip rather than seven.
NFR-05	Read-only endpoints open no transaction, which saves two round trips to a remote database.
NFR-06	The front desk badge asks for a count only (`?brief=1`), which runs one aggregate query.
NFR-07	The job loop first checks, in one query, which jobs are due. A pass with nothing due costs one query.

2.3 Usability

ID	Requirement
NFR-11	Every screen works at a width of 375 px.
NFR-12	A refusal tells the user what to do instead, as a sentence rather than a code.
NFR-13	Text renders Vietnamese diacritics correctly, and currency is never upper-cased (1,500,000đ, never 1,500,000Đ).
NFR-14	Motion is decorative and never carries meaning on its own.

2.4 Auditability

ID	Requirement
NFR-15	Every staff-only change writes an audit row naming the actor, action, record and time. This covers refunds, reconciliation, shift closes, settings, sessions, classes, staff accounts, attendance corrections and payment links. No endpoint deletes an audit row.
NFR-16	Member, payment and invoice codes come from a counter table and have no gaps within each kind.

2.5 Security and privacy

Authorisation is decided on the server for every request, and member data stays inside the centre.

ID	Requirement
NFR-17	Passwords are stored only as salted hashes and are never written to logs.
NFR-18	Session tokens are stored as SHA-256 hashes. Member sessions last seven days and staff sessions twelve hours.
NFR-19	Every request is authorised on the server. Hiding a button is not a control.
NFR-20	A member can read only their own bookings, receipts and profile. A read of another member's data returns `403`.
NFR-21	Health notes and guardian details are shown only to those who need them and are never sent to the assistant's model.
NFR-22	In production, one-tap demo sign-in and on-screen codes are off unless `VITE_DEMO_LOGINS` or `OTP_ECHO` is set explicitly.
NFR-23	A switched-off module (F4 training) answers `403` with `BR-62` on every route.
NFR-24	Report exports are built on request from the same queries as the screens, and are not stored.
NFR-25	An online payment is trusted only when read from payOS or carried by a verified webhook signature.
NFR-26	Sign-in locks for 15 minutes after five wrong passwords, and a one-time code after five wrong attempts.

2.6 Maintainability

The requirements in this document are checked by executable tests, so a change that breaks one fails before release.

- `npm test` runs the unit tests for pricing, refunds, check-in windows and other rules.
- `npm run check:api` drives the real API against a real database and checks the money and schedule rules end to end.
- `npm run typecheck` checks the TypeScript types across the application.

V. Requirement appendix

This appendix lists every business rule, configurable setting, shared requirement and user message that the sections above refer to. Each entry matches the running application.

1. Business rules

A refused request returns `422 BR_VIOLATION` with the rule code in `br`, so any refusal on screen traces to one row below.

ID	Rule	Applies to
BR-01	One account per phone number or email.	UC-02
BR-02	A password has at least 8 characters, with letters and digits.	UC-02, password change
BR-07	A member under 16 needs a guardian's name and phone.	UC-02
BR-08	The terms and data-privacy notice must be accepted.	UC-02
BR-12	Class enrolment needs an active plan covering the sport.	UC-24
BR-14	A frozen plan must be unfrozen before it is frozen again or paid for.	Freeze, plan payment
BR-15	A plan may be frozen for at most 30 days a year.	Freeze
BR-17	Paying for a court from a plan needs court hours left.	Booking by plan hours
BR-18	A plan's session quota must not be used up.	UC-24
BR-19A	A plan grants benefits only between its start and end dates.	Pricing
BR-20	A class enrolment can be cancelled up to 4 hours before the next session.	Cancel enrolment
BR-21	Moving a session or changing its coach is refused when the court or coach is busy.	Class changes
BR-22	A class cannot be overfilled, and capacity cannot drop below those enrolled.	UC-24, class edit
BR-23	The coach must be active and teach the class sport.	Class setup
BR-24	No duplicate or clashing enrolment.	UC-24
BR-25	A waitlist offer expires after 2 hours.	UC-25
BR-26	When the centre cancels a session, a per-session plan gets that session back once.	Session cancel
BR-27	A class can be handed to a substitute coach from now on.	Change coach
BR-28	Moving a session less than 12 hours before its old time needs a reason.	Session move
BR-32	Courts are booked at most 7 days ahead, and at most 2 slots a day.	UC-12
BR-34A	Plan discount percentages are whole numbers from 0 to 100.	Plan setup
BR-34B	With no matching price rule, the system refuses to quote and never invents a price.	UC-12
BR-35	Bookings fall inside opening hours, on a court that is ready.	UC-12
BR-38	Gear rental cannot exceed the stock on hand.	Gear
BR-39	One court time has exactly one owner (see NFR-01).	All bookings
BR-39B	Court check-in opens 15 minutes before the start and closes 10 minutes after.	Court check-in
BR-39C	A slot that clashes with one of the member's classes needs confirmation.	UC-12
BR-39H	A replacement hold is checked in full before it replaces the old one.	UC-12

ID	Rule	Applies to
BR-42A	Prices include VAT, the only display mode in this version.	Pricing
BR-43A	Money is rounded once, half up, after the discount.	Pricing
BR-44	A promo code must be active, within its dates, and fit the order's scope, sport, plan and minimum.	UC-22
BR-45	A promo code cannot be used beyond its total or per-member limit.	UC-22
BR-46	A code that does not stack never adds to a plan discount; the better one applies.	UC-22
BR-49	Taking money at the desk needs an open till shift.	UC-44
BR-53	A class register locks 2 hours after the session; only a manager can correct it, with a reason.	Attendance
BR-54	Gate check-in never fills class attendance.	UC-46
BR-55	A student's health notes are visible only within the viewer's scope.	Coach portal
BR-56	Training records never charge money or change a plan balance.	Coach portal
BR-57	Late is the coach's mark in the register.	Attendance
BR-58	Three absences in a row alert the coach and managers; an excused absence breaks the streak.	Attendance
BR-59	A coach works only on classes they lead or assist.	Coach portal
BR-62	A switched-off module answers `403` with `BR-62`.	Training module (F4)
BR-65	A withdrawn plan cannot be ordered.	Plan orders
BR-66	No booking in the past; a walk-in needs at least 20 minutes left in the slot.	UC-12, walk-in
BR-67	A class must be open to enrol.	UC-24
BR-70	A promo code that has been used keeps its name; staff pause it and create a new one.	Promo codes
BR-71	A gate QR code must be a valid member code that has not been used.	UC-46
BR-72	A member with nothing to admit on needs an override reason, and the visit is flagged. A repeat scan within 120 minutes returns the first visit.	UC-46
BR-73	Gate check-in never spends a session or changes a balance.	UC-46
BR-75	A training session with results cannot change its plan; the coach copies it instead.	Coach portal

The member debt limit and court-hour conversion were removed in v1.4.1. Code BR-44 now covers promo code validity.

2. Centre settings

Managers change these values on Centre settings (`/manager/settings`), and the rules above read them on every request.

Group	Setting	Default
Locale	Time zone; currency	Asia/Ho_Chi_Minh; VND (both fixed)
Opening hours	Open; close; slot length	06:00; 22:00; 60 minutes
Booking	Hold time; days ahead; slots per member per day	5 minutes; 7 days; 2
Cancellation	Court booking notice; class notice	2 hours; 4 hours
Court refunds by notice	24 hours or more; 4 to 24 hours; under 4 hours	100%; 50%; 0%

Group	Setting	Default
Check-in	Court check-in window; no-show grace	15 minutes before; 10 minutes after
Deposit	Deposit share; peak slots only	30%; yes
Fixed weekly bookings	Length; discount	4 to 12 weeks; 10%
Money	Refund limit without a manager; VAT rate; rounding	1,000,000đ; 0%; nearest 1,000đ
Members	Freeze days a year; minor age; waitlist offer	30 days; under 16; 2 hours
Transfers	Bank transfer hold	120 minutes
Loyalty and day pass	Spend per point; value per point; day pass price	10,000đ; 100đ; 60,000đ
Gate	Self check-in; repeat-scan window	Off; 120 minutes

3. Common requirements

These requirements apply to every function, so the use cases do not repeat them.

Area	Requirement
Idempotency	Requests that take money or change a schedule accept an `Idempotency-Key`; a replay returns the first result.
Audit	Every staff-only change writes an audit row, and no endpoint deletes one.
Money	Amounts are whole VND, rounded once to the nearest 1,000đ after discounts. Payments are never deleted; a refund is a new negative payment.
Codes	Member, payment and invoice codes have no gaps within each kind.
Time	All times are stored with a time zone and shown in Asia/Ho_Chi_Minh time.
Sessions	Member sessions last seven days and staff sessions twelve hours.
Lockout	Five wrong passwords, or five wrong one-time codes, lock that path for 15 minutes.
Rate limits	Assistant questions: 800 characters each, 20 a minute. Plan orders: 3 a minute, 10 an hour.
Notifications	Notifications are queued and sent in batches of 50, with up to five attempts each. Plan reminders go 7, 3 and 0 days before the end date.
Background jobs	The jobs listed in III.1.4 run in one loop every 15 seconds.
Language	Every screen is available in English and Vietnamese, and Vietnamese diacritics render correctly.
Layout	Every screen works at a width of 375 px.

4. Application messages

Each message below is the text the application shows today. The code travels alongside it so support can trace the refusal.

#	Code	Type	Message
MSG01	BR-01	Error	That phone or email already has an account.
MSG02	BR-02	Error	Password needs at least 8 characters, with letters and numbers.
MSG03	RATE_LIMITED	Error	Too many OTP attempts — locked for 15 minutes.
MSG04	RATE_LIMITED	Error	5 wrong passwords — locked for 15 minutes.
MSG05	BR-32	Error	You can book at most 7 days ahead.

#	Code	Type	Message
MSG06	BR-39C	Confirmation	This slot clashes with a class you are in. Confirm to hold it anyway.
MSG07	CONFLICT_SLOT	Error	Someone just took that slot.
MSG08	HOLD_EXPIRED	Error	Your hold has expired.
MSG09	BR-34B	Error	There is no price for that slot yet — ask the desk.
MSG10	BR-44	Error	That code has expired.
MSG11	BR-12	Error	You need an active plan covering this sport.
MSG12	BR-25	Error	That offer has expired.
MSG13	BR-49	Error	Open a till shift before taking payment.
MSG14	BR-71	Error	That code was already used — ask the member to refresh it.
MSG15	VALIDATION	Error	This payment has already been refunded in full.
MSG16	FORBIDDEN	Error	You raised this refund, so someone else has to sign it off.
MSG17	CONFLICT_STATE	Error	That refund is no longer waiting for a decision.
MSG18	BR-53	Error	This register closed 2 hours after the session. Ask a manager to correct it.
MSG19	FORBIDDEN	Error	You do not have permission.
MSG20	RATE_LIMITED	Error	Too many attempts — try again later.